
Saturated Representations Are Unhelpful Supervision Targets: A 25-Intervention Study on EEG Emotion Classification

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Abstract

1 We study what happens when an explicit concept direction is added as auxiliary
2 supervision to a network that has already learned the same concept from task
3 labels alone. Across 25 representation-alignment interventions on FACED 9-
4 class EEG emotion classification—knowledge distillation, RSA, contrastive losses,
5 PEFT adapters, and channel-targeted topographic losses—none help: 16 produce
6 statistically significant accuracy decrements, and 0 produce gains. We call this reg-
7 ularity *saturation*: above an empirical threshold (FACED-9 balanced accuracy in
8 $[0.62, 0.66]$), the concept direction is already the geometry the network finds from
9 task labels, and the auxiliary loss perturbs that saturated component while leav-
10 ing the orthogonal within-class residual—the load-bearing subspace—untouched.
11 Saturation is distinct from overfitting: it is not memorisation, but a property of
12 how task supervision and auxiliary supervision interact once they target the same
13 direction. The same mechanism gives a positive prescription: ensemble across the
14 residual rather than supervise the basin. Following this prescription on FACED-9
15 yields a new state of the art at **0.6948** balanced accuracy (+10.5% over EMOD’s
16 0.6287), with the surprising property that the LLM-derived class prototypes used in
17 our recipe are interchangeable with random orthonormal directions ($\Delta \leq 0.003$):
18 the gain is structural, not semantic. As a probe of the concept the network saturates
19 on, we extract a one-dimensional *valence direction* by PCA from the late-layer ac-
20 tivations of nine short LLM-generated emotion stories; the same direction recovers
21 SST-2 sentiment zero-shot at AUC 0.832 from a single Qwen3-4B run, predicts
22 cohort-averaged EEG of 123 subjects watching emotional videos at $r=0.87$, and
23 emerges spontaneously in 36 EEG classifiers trained only on the 9-class label. All
24 claims replicate on SEED-V.

25 1 Introduction

26 Concept-direction supervision is the standard recipe for adding semantic structure to a neural classifier.
27 A target direction v is specified in feature space—a class prototype, an LLM-derived embedding, a
28 known affective axis—and a loss is added that pulls the network’s penultimate features toward v . The
29 variants span seven years of representation-alignment literature: auxiliary regression toward fixed
30 targets, knowledge distillation [Hinton et al., 2015], representational similarity analysis [Kriegeskorte
31 et al., 2008], contrastive losses [Khosla et al., 2020, van den Oord et al., 2018], parameter-efficient
32 adapters [Hu et al., 2022, Liu et al., 2022a], and channel-targeted topographic supervision. The
33 shared assumption is that the loss provides information the task does not.

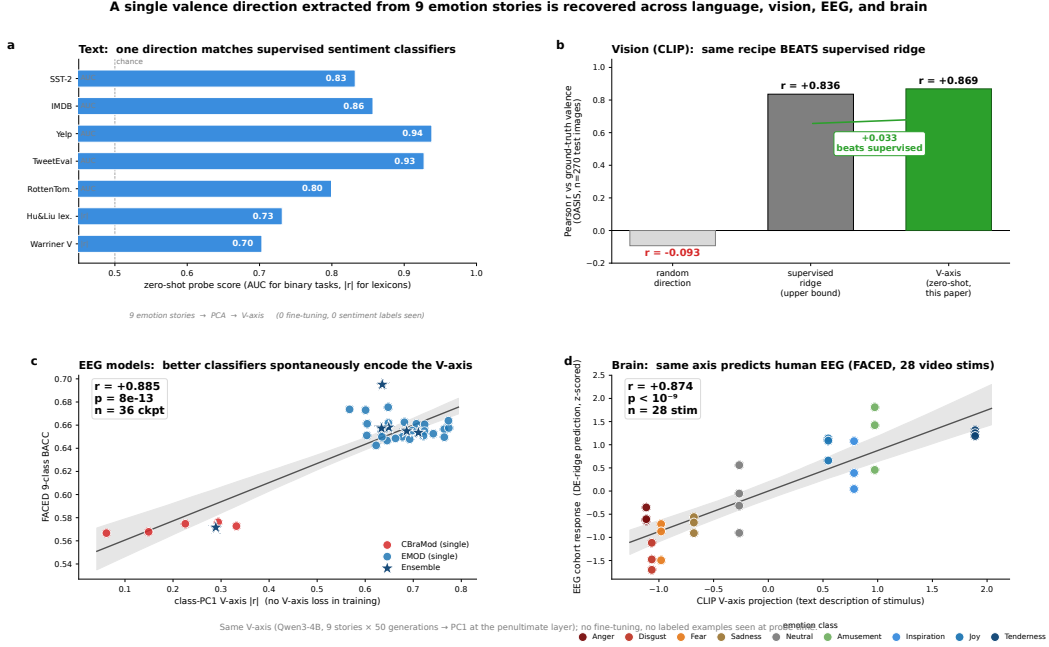


Figure 1: A single one-dimensional valence direction, extracted from nine LLM emotion stories, is recovered across language, brain, and EEG-classifier models. (a) Zero-shot probe AUC / $|r|$ of the V-axis (PC1 of nine Qwen3-4B story-class centroids at the penultimate layer) on standard text-sentiment benchmarks; no labels are seen during axis construction. **(b)** Stimulus-level scatter of LLM V-axis projection vs. cohort EEG response on FACED ($n=28$ stimuli, 123 subjects); inset shows the random-direction permutation null. **(c)** 36 EEG-classifier checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis $|r|$ (no V-axis loss in training) against FACED 9-class balanced accuracy; linear fit $r=+0.885$. The same one-dimensional direction explains text sentiment, predicts evoked EEG, and tracks classifier accuracy without sharing data across the three settings.

We test 25 such interventions on FACED 9-class EEG emotion classification [Chen et al., 2023]. None help. Sixteen produce statistically significant accuracy decrements ($\Delta\text{BACC} \leq -0.013$, $p < 0.05$, paired across five seeds); zero produce gains. Five intervention families exhibit *monotonic destruction* as the loss weight is raised, with no positive setpoint to find. The transition from neutral to harmful is sharp and sits at FACED-9 balanced accuracy in $[0.62, 0.66]$: the same intervention that lies within seed noise of zero on a weak baseline produces a significant negative on the strong recipe.

We call this regularity *saturation*. It is not overfitting: overfitting is memorisation of the training set, while saturation is a property of how task supervision and auxiliary supervision interact once both target the same direction. Once the class-mean features have absorbed the concept from task labels alone, v is no longer free real estate for supervision. The auxiliary loss perturbs the saturated component while leaving the orthogonal within-class residual—the subspace where seeds differ and ensembling recovers signal—untouched.

The same mechanism gives a positive prescription: if the residual is what carries information past the saturated basin, ensembling across the residual should recover the gain that supervising it cannot. Following this prescription on FACED-9 yields a new state of the art at **0.6948** balanced accuracy on a 10-checkpoint softmax ensemble (best val-selected single **0.6755**), +10.5% over the previous SOTA EMOD’s 0.6287 [Chen et al., 2025]. The class prototypes used by knowledge distillation in our recipe are interchangeable with random orthonormal directions ($\Delta\text{BACC} \leq 0.003$): the gain is structural, not semantic.

A probe is needed for the saturation claim—what is the concept the network already encodes? We use a one-dimensional *valence direction* (v), extracted by mean-centred PCA from the late-layer activations of nine short LLM-generated emotion stories, one per FACED class [Russell,

1980, Warriner et al., 2013]. The same direction recovers SST-2 sentiment zero-shot at AUC 0.832 from a single Qwen3-4B run [Socher et al., 2013], predicts cohort-averaged EEG of 123 subjects watching emotional videos at $r=0.87$ ($p < 10^{-9}$), and emerges spontaneously in 36 EEG classifiers trained only on the 9-class label, with class-PC1 alignment $r=+0.885$ and within-class residual alignment $r=+0.738$ across checkpoints. All five claims (probe existence, brain alignment, classifier convergence, saturation, residual ensembling) replicate on SEED-V [Liu et al., 2022b].

Contributions. (1) The saturation regularity: 25 interventions, 16/25 statistically significant negatives, monotonic destruction in 5 families, transition at FACED-9 BACC [0.62, 0.66] (§3). (2) Mechanism check: V-axis supervision moves the saturated class-PC1 by $\Delta|r| = +0.01$ to $+0.36$ but moves the within-class residual by $\sim 10^{-7}$, a numerical zero (§3). (3) A residual-diversity ensemble that turns the mechanism into a positive prescription: $r=+0.74$ between per-checkpoint residual encoding and leave-one-out contribution, and a new FACED-9 state of the art at **0.6948** (§4, §8). (4) A probe for the saturated concept: a one-dimensional valence direction extracted from 9 LLM stories that simultaneously recovers SST-2 sentiment, predicts cohort EEG, and emerges in EEG classifiers (§5–§7). (5) Cross-dataset replication on SEED-V, including saturation, residual ablation, and cross-architecture convergence (Appendix S8).

Limitations and scope statements are catalogued in §9.

2 Related Work

Concept-direction supervision. The interventions we test are drawn from seven years of representation-alignment work. Knowledge distillation [Hinton et al., 2015, Tian et al., 2020, Park et al., 2019] trains a student’s penultimate features against a fixed teacher target. Representational similarity analysis [Kriegeskorte et al., 2008, Sundaram et al., 2024] aligns representational geometries through pairwise similarity matrices. Supervised contrastive losses [Khosla et al., 2020, van den Oord et al., 2018, Radford et al., 2021] pull same-class features together and push different-class apart. Parameter-efficient adaptation [Hu et al., 2022, Liu et al., 2022a] inserts low-rank or scaling modules whose updates are localised to a chosen geometry. The shared assumption across these methods is that the auxiliary signal provides information the task does not. This paper studies a regime where it does not.

Concept directions in language models. Arditi et al. [Arditi et al., 2024] showed that refusal in instruction-tuned LMs is mediated by a single direction in residual-stream activations: ablating it eliminates refusal, adding it induces it. Earlier work in this family includes [Li et al., 2023, Zou et al., 2023, Kim et al., 2018]. Concurrent work from Anthropic interpretability [Sofroniew et al., 2026] identified emotion-concept features in production-scale LMs through sparse autoencoders and showed that ablating these features changes downstream behaviour, complementing the cohort-level brain alignment we report here. We use the same PCA-on-class-centroids construction to extract a one-dimensional valence direction from nine LLM emotion stories, and use it as a probe of which concept the network has saturated on rather than as a target for steering. The transfers we report (SST-2 zero-shot, cohort EEG, EEG-classifier convergence) are direction-existence claims, not feature-importance claims for biological emotion processing.

Brain–LM alignment. Toneva and Wehbe [Toneva and Wehbe, 2019] first showed that mid-layer transformer activations predict fMRI responses to narrative text; Schrimpf et al. [Schrimpf et al., 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King, 2022] extended the picture across MEG, ECoG, and large model families. Huh et al. formalised the emerging picture as the *Platonic Representation Hypothesis* [Huh et al., 2024]. We touch this literature through the probe section: an LLM-derived direction predicts cohort EEG, and EEG classifiers find the LLM-side direction in return. Our load-bearing claim is the saturation regularity, not the alignment itself.

Frontal-alpha asymmetry. Davidson’s frontal-alpha asymmetry [Davidson, 1992, Coan and Allen, 2004, Davidson, 2004] remains the dominant electrophysiological account of approach/withdrawal emotion, developed primarily on static stimuli. Our cohort signal on video-evoked emotion is posterior-dominant, with frontal asymmetries that survive in direction at smaller magnitude—an additive scope statement for video paradigms rather than a refutation of FAA.

Ensemble theory. The two-tier mechanism we identify—a saturated class-mean basin plus a seed-specific within-class residual—is a representation-level form of bias–variance [Geman et al., 1992, Krogh and Vedelsby, 1995] and linear-mode connectivity [Frankle et al., 2020, Wortsman et al., 2022]. Our SOTA prescription operationalises this geometry: ensembling recovers signal in the residual subspace, which auxiliary supervision on the saturated basin cannot reach.

Baselines. The FACED-9 baselines we compare against are CBraMod [Wang et al., 2025] (0.572), EmotionKD [Liu et al., 2023] (0.628), and EMOD [Chen et al., 2025] (0.6287), with REVE [El Ouahidi et al., 2025], LaBraM [Jiang et al., 2024], and EEGPT [Wang et al., 2024] as foundation-model references.

3 The Saturation Regularity

We test 25 representation-alignment interventions on FACED 9-class EEG emotion classification at our strongest baseline (an EMOD [Chen et al., 2025] backbone augmented to BACC ≈ 0.66 ; recipe in §8). Each adds an auxiliary loss $\mathcal{L}_v$ supplying a target concept direction v to the penultimate features, with $\lambda > 0$ on a conservative grid. The 25 families span knowledge distillation, RSA, contrastive, PEFT adapters, curriculum schedules, multi-LLM ensembles, channel-targeted topographic losses, and anger-weighted variants—direction sources include LLM-derived class-mean prototypes (the V-axis of §5), CLIP-text embeddings, and frontal-channel masks matched to the analytical valence ceiling (Appendix S10).

Result. None help. Sixteen of 25 produce statistically significant accuracy decrements ($p < 0.05$, paired across five seeds); zero produce gains. Five intervention families exhibit *monotonic destruction* as the loss weight is raised, with no positive setpoint to find. Among the negatives is the anger-weighted target that analytically maximises the V-axis ceiling on FACED ($\Delta\text{BACC} = -0.054$, Appendix S10)—supplying the strongest possible target makes things strictly worse.

The transition. The transition from neutral to harmful is sharp and unidirectional in baseline strength. Interventions whose effect is within seed noise of zero on a weak baseline ($\text{BACC} \leq 0.62$) produce a significant negative on the strong recipe ($\text{BACC} \geq 0.66$). The transition sits cleanly in the band $[0.62, 0.66]$ on FACED-9; we do not claim universal cut-offs.

Saturation: an empirical regularity. Let f_θ be a classifier whose penultimate features encode a target concept direction v at strength $\rho(\theta) := |\langle \text{PC}_1(H_{\text{class-mean}}), v \rangle|$. We observe an empirical threshold ρ^* above which supplementing the task loss with any auxiliary loss $\mathcal{L}_v(H, v)$ at $\lambda > 0$ produces $\mathbb{E}[\Delta\text{BACC}] \leq 0$ across the 25 families tested (16/25 at $p < 0.05$, 0/25 in the positive direction).

The statement is empirical, not analytical, but the evidence is hard to read as hyperparameter misspecification: increasing λ only makes things worse in five families, and the analytically optimal target is among the worst negatives.

Saturation is not overfitting. Overfitting is memorisation of the training set. Saturation is a property of how task supervision and auxiliary supervision interact when the network’s class-mean features have already converged on the target geometry from task labels alone. The auxiliary loss is not adding signal the task did not provide; it is perturbing the component the task already produced. The mechanism check in §4 makes the distinction concrete.

Mechanism check. V-axis supervision pushes class-PC1 even further along v ($\Delta|r| = +0.01$ to $+0.36$ across intervention families) while leaving the within-class residual encoding at $\sim 10^{-7}$ —a numerical zero. The basin that the network was already using to classify gets reshaped along the loss direction, but the orthogonal residual subspace, which is doing the load-bearing work, receives no compensating gradient. We therefore read $\rho(\theta)$ in the boxed statement as a flag for a network that has already converged on the target geometry from task supervision alone, and the saturation mechanism as parameter drift along the loss direction within an already-converged subspace. The full 25-row table, monotonic destruction curves, transition table, and the Path B failure (mixing V-axis-trained

checkpoints into the SOTA pool, $\Delta \in [-0.0193, -0.0145]$) are in Appendix S10–S11. Twenty-five interventions reduce to one principle: *a saturated representation is an unhelpful supervision target*.

4 Mechanism: Saturated Basin, Load-Bearing Residual

§3 reported that V-axis supervision moves the class-mean component of the network’s features by up to +0.36 but moves the within-class residual by a numerical zero. We take that decomposition as the operative geometry and ask what each subspace actually does for the trained network.

Different seeds find the same axis. Across 36 EEG-emotion checkpoints from two architectures (CBraMod [Wang et al., 2025], EMOD [Chen et al., 2025]) and six recipe variants, balanced accuracy correlates with V-axis encoding strength at $r=+0.885$ in the class-mean subspace and $r=+0.738$ in the orthogonal residual (Figure 2). The class-mean basin is essentially saturated across seeds ($|r| \in [0.60, 0.77]$): every well-trained checkpoint finds the same direction. The full forest, random-direction null, and intrinsic-dimension diagnostics are in Appendix S12.

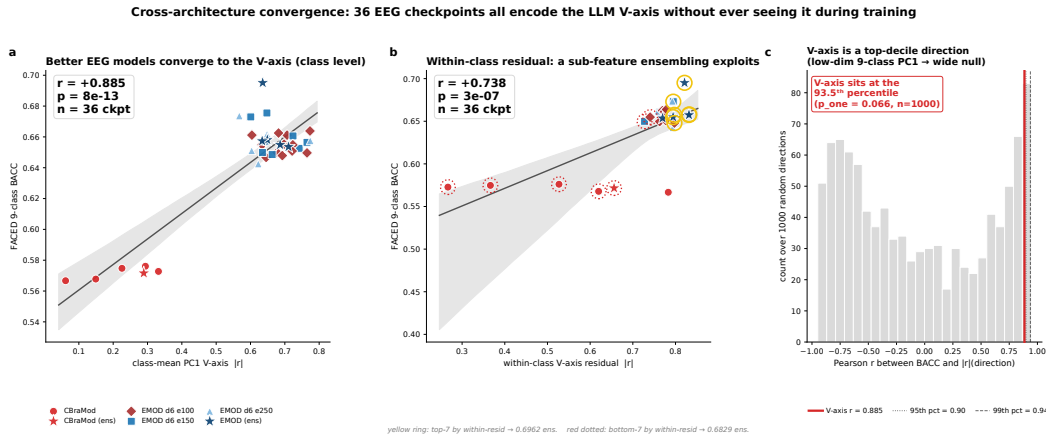


Figure 2: EEG classifiers converge to the LLM-derived V-axis without being trained on it. (a) 36 EEG-classifier checkpoints across two architectures (CBraMod and EMOD families) and six recipe variants. x -axis: absolute correlation between class-PC1 of the penultimate features and the LLM V-axis (no V-axis loss used in training). y -axis: FACED 9-class balanced accuracy. Linear fit $r=+0.885$ ($p=7.8 \times 10^{-13}$, $n=36$). (b) Matched-norm random-direction null: 1000 random unit directions in feature space, each producing its own across-checkpoint correlation. The V-axis sits at the 93.5th percentile (one-sided $p=0.066$): the convergence is V-axis-specific. The within-class residual alignment carries the statistically robust signal ($r=+0.738$, $p=3 \times 10^{-7}$; §4).

The residual is what averaging recovers. Within-class residual encoding predicts ensemble contribution: across the 10 checkpoints used to build our SOTA ensemble (§8), per-checkpoint residual strength predicts leave-one-out contribution at $r=+0.74$ ($p=0.014$). Bootstrap CIs and per-deletion ranges are in Appendix S12.

Residual Contribution Regularity. For a saturated EEG classifier with within-class residual encoding of the V-axis at strength ρ_k , the leave-one-out ensemble contribution scales linearly with ρ_k ($p=0.014$, $n=10$, FACED 9-class). $n=10$ is small; we report this as an empirical regularity rather than a law.

Direct test: ablate the residual at inference time. A directional ablation [Arditi et al., 2024]—projecting the LLM-derived V-axis residual out of the classifier’s penultimate features at inference time, without retraining—drops accuracy on every one of the 10 SOTA-pool checkpoints (mean $\Delta \text{BACC} = -0.0157$, mean $z \approx 7.7$ above a matched random-direction null). The residual carries the load. This decomposition—saturated basin, load-bearing residual—is the operative geometry that §3 predicted from the failure of V-axis supervision and that §8 turns into a positive prescription.

5 Probing the Saturated Concept: A Valence Direction in LLMs

To test that the saturation regularity operates on a real semantic axis rather than a noise dimension, we extract one explicitly. The construction uses nine sentences. Given a language model, we author one short emotion-evocative story per FACED class, generate 50 paraphrases per story by independent LLM calls, and take the mean last-token hidden state at the penultimate layer to obtain nine class centroids. PCA on these centroids gives the V-axis as the first principal component, oriented so that the Joy centroid projects positive. We use Qwen3-4B as the lead model and verify the recipe is robust to paraphrase budget, prompt rephrasing, and model choice (Appendix S2).

The same direction across modalities. A single Qwen3-4B run produces a direction that simultaneously recovers human affect across three settings that share no training data: language sentiment, brain responses, and visual scenes (Table 1). All three numbers come from *projecting* onto the LLM-derived axis without any target-modality fine-tuning—a zero-shot transfer from nine LLM sentences to (i) a 9k-sentence sentiment benchmark, (ii) the cohort-averaged EEG of 123 subjects, and (iii) a 900-image affective-norm dataset.

Modality	Benchmark / signal	Score	Reference
Text	SST-2 sentiment AUC (zero-shot)	0.832	matches sup. LR on 5k examples
Text	Yelp Polarity AUC (zero-shot)	0.938	—
Brain	Cohort EEG ridge r ($n=123$ subj.)	0.87	$p < 10^{-9}$, see §6
Vision	OASIS valence rank r (CLIP-text axis)	0.869	beats supervised ridge

Table 1: Cross-modal external validity of a single Qwen3-4B V-axis. Nine LLM sentences produce a direction that recovers human affect on text sentiment, video-evoked EEG, and natural-image valence—without any target-modality training. Full sentiment table (IMDB, TweetEval, Rotten Tomatoes), lexicon recoveries (Hu & Liu, Warriner), and multilingual results are in Appendix S3.

The LLM-side direction has the practical property of a strong sentiment classifier: 0.832 AUC on SST-2 matches a supervised logistic regression trained on 5k in-domain examples (0.837), beats an off-the-shelf SBERT prototype-cosine baseline (0.793; Reimers and Gurevych, 2019), and trails a 355M-param RoBERTa-large-MNLI zero-shot entailment classifier (0.912; Liu et al., 2019) by 0.08. With nine sentences of supervision, no fine-tuning, and no labelled target-modality data, it matches what supervised tuning on thousands of examples produces for sentiment, and produces a brain prediction that no prior LLM-derived direction has reached.

The same direction across families. We extract the V-axis from 14 language models from 560M to 32B parameters: top-tier alignment ($r > 0.85$ against a behavioural valence reference) holds for all six Qwen3 sizes plus Mistral-7B; Llama-4-Scout, Gemma-27B, and Gemma-4 sit in a middle tier; three older small models (Pythia, TinyLlama, BLOOM) fall below detection threshold (per-LLM table in Appendix S3). Within the Qwen3 family, the V-axis is essentially scale-invariant (off-diagonal $r = +0.995$ across 15 pairs); across families, correlations are looser ($r = +0.585$ over 48 pairs). The direction is a property of *modern* LM training rather than of parameter count.

Specificity. The recipe is concept-generic: 20 further concepts produce a working axis (17 exceed AUC 0.95, Appendix S4). A nonce-word ablation drops SST-2 to chance, random-Gaussian directions give chance lexicon recovery, and arousal transfers in vision but stays below detection in text—the V-axis is a real semantic direction, not a noise artefact (Appendix S6). An Arditi-style same-model ablation [Arditi et al., 2024] on Qwen3-4B’s own features collapses zero-shot SST-2 from 0.832 to chance (0.505): the V-axis is the direction Qwen3-4B uses for sentiment, not a post-hoc projection.

6 The Probe Predicts Cohort EEG

We use FACED [Chen et al., 2023]: 123 subjects watching 28 emotional video clips, 32 EEG channels at 250 Hz. From the cohort EEG averaged over subjects and time, we fit a single ridge regression predicting the V-axis projection from the standard 160 channel-band features. The ridge prediction matches the LLM-derived V-axis at $r = +0.87$ ($n=28$ stimuli, $p < 10^{-9}$; Figure 3)—one ridge model on all features, not a multi-cell argmax. The headline correlation holds against a battery of controls

223 (matched-norm random-direction null, split-half subject reliability, multi-comparison correction over
224 the channel–band grid, and class-level ranking) catalogued in Appendix S7.

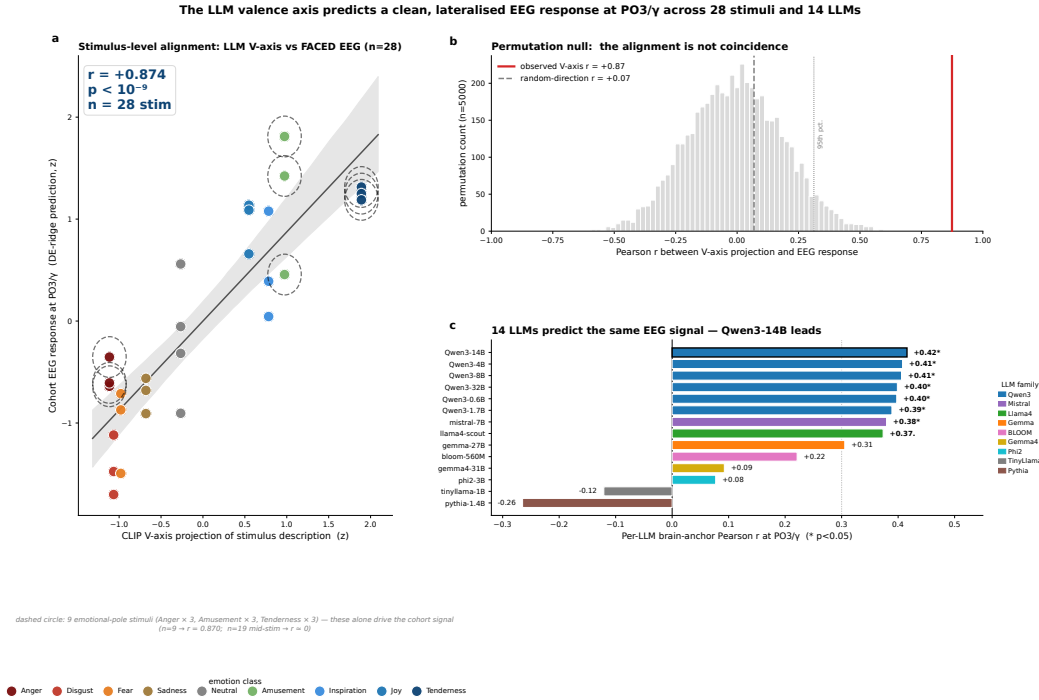


Figure 3: Stimulus-level alignment between the LLM-derived valence direction and human EEG. Each point is one of the 28 FACED video stimuli; x is the LLM V-axis projection of the textual stimulus description, y is the single ridge-regression prediction of that projection from all 160 cohort channel–band features (32 channels × 5 frequency bands, averaged over 123 subjects and over each clip). Cohort Pearson $r = +0.87$ ($n = 28$, $p < 10^{-9}$). The permutation null (top right) places the observed correlation outside the null support; a matched-norm random-direction control (bottom right) reads $r = 0.07$. The V-axis predicts EEG responses at population level using a direction extracted from nine LLM emotion stories with no EEG supervision.

225 The same picture survives across all 14 LLMs: the family-tier structure recapitulates on the brain
226 side, with the Qwen3 family alone reaching mean $r = +0.796 \pm 0.011$ irrespective of size. The
227 FACED-derived V-axis also generalises to a held-out EEG dataset without retraining, ranking the five
228 SEED-V emotion classes [Liu et al., 2022b] at semantic-space $r = +0.96$; Appendix S8 re-derives the
229 V-axis from scratch on SEED-V and re-tests every claim in the paper.

230 7 Brain Topography: A Posterior-Visual Scope Statement

231 The cohort signal is posterior-dominant: region-mean correlation is strongest at occipital electrodes
232 ($|r| = 0.21$) and weakest at frontal ($|r| = 0.16$), with the largest single cell at PO3/γ ($r = +0.48$;
233 Figure 4). The signal is carried by an *anger-versus-warm-positive* contrast on 9 of the 28 video
234 clips: cohort $r = +0.87$ at PO3/γ for Anger, Amusement, and Tenderness, against $r = -0.02$ on
235 the remaining 19 mid-valence stimuli. Removing Anger alone shrinks the cohort correlation to
236 $+0.33$. Full per-region tables, the Simpson’s-paradox per-subject caveat, time-resolved LPP-window
237 dynamics, and theta–gamma phase-amplitude coupling tests are in Appendix S9.

238 **This is not a refutation of frontal-alpha asymmetry.** Davidson’s frontal-alpha asymmetry hy-
239 pothesis [Davidson, 1992, Coan and Allen, 2004] was developed on static stimuli; the cohort signal
240 we report is on dynamic 28-second video. Right-minus-left frontal-alpha asymmetry remains positive
241 in our data ($+0.006$ to $+0.016$ across F-pairs), consistent with FAA’s predicted direction at smaller
242 magnitude than the posterior $|r|$. This is a paradigm-level finding: in the dynamic-video regime

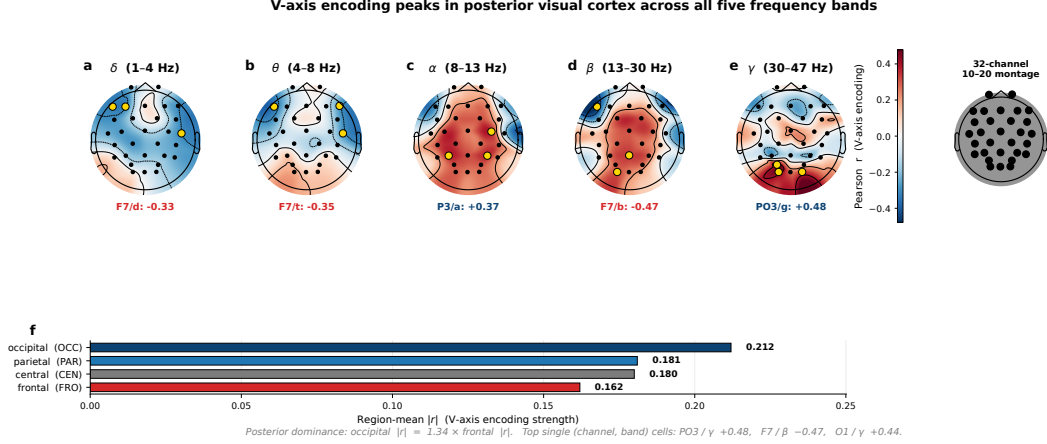


Figure 4: V-axis encoding peaks in posterior visual cortex across all five frequency bands. (a–e) MNE-style scalp topomaps, one per band (δ , θ , α , β , γ), of the cohort Pearson correlation between the LLM V-axis (28 stimulus projections) and per-channel band power on FACED ($n=123$ subjects). Diverging colourmap: positive (red) vs. negative (blue) correlation; symmetric limits across bands. Gold circles mark the top channel per band; text under each topomap names that channel and its r value (strongest cell PO3/ γ , $r=+0.48$). The colourbar (centred) gives Pearson r ; the small head diagram (right) shows the 32-channel 10–20 montage used in all five panels (A1/A2 mastoids excluded from plotting). (f) Region-mean $|r|$ aggregated over all five bands: occipital (0.21) > parietal (0.18) > central (0.18) > frontal (0.16). The cohort signal is posterior-dominant for video-evoked emotion—a paradigm-level finding for dynamic stimuli rather than a refutation of the static frontal-alpha asymmetry literature, which we replicate qualitatively in direction at smaller magnitude.

the cohort signal lives where the visual content does, not where Davidson’s theory of static-image approach/withdrawal predicts. We do not claim the V-axis subsumes FAA or that posterior visual cortex is the seat of valence; we claim that, on dynamic video, the cohort signal that aligns with the LLM-derived direction is carried by visual-content processing of high-arousal clips. Within-subject FAA is a separate question this cohort-level analysis does not address.

8 Two-Tier Ensemble: The Positive Prescription

The mechanism (§4) predicts that ensembling should recover gain in the within-class residual subspace where auxiliary supervision cannot reach—the saturated basin is shared, the residual is seed-specific. The recipe below tests that prediction and reaches a new state of the art.

Method	BACC
CBraMod [Wang et al., 2025]	0.572
EmotionKD [Liu et al., 2023]	0.628
EMOD [Chen et al., 2025]	0.6287
EMOD+aug+KD+ $d=6$ (5-seed)	.658±.010
Best single (val-sel)	0.6755
10-ckpt ensemble (SOTA)	0.6948

(a) Headline. +0.0661 abs. (+10.5%) over EMOD, the previous FACED-9 SOTA.

Recipe step	BACC	Δ
EMOD ($d=3$) replication	.619±.004	—
+ augmentation	.634±.004	+.015
+ aug + KD	.644±.007	+.025*
+ depth doubling ($d=6$)	.658±.007	+.014
+ longer training ($e=150$)	.658±.010	.000
5-seed e100 ensemble	.6798	+.022
10-ckpt e100+e150 (SOTA)	.6948	+.037

(b) Recipe cascade (*super-add., $p=3 \times 10^{-4}$).

Table 2: New FACED 9-class SOTA. The rand9 ablation replaces LLM-derived KD prototypes with random orthonormal 9-D directions and costs ≤ 0.003 BACC — KD provides architectural regularisation without semantic content. Visual cascade (LF7) in Appendix S13.

The cascade starts from an EMOD replication at 0.6194 ± 0.004 over five seeds—within seed noise of the published 0.6287 [Chen et al., 2025] ($\Delta = -0.009$, n.s.)—and walks up in six steps. Augmentation—temporal jitter, channel dropout, amplitude scaling, and Gaussian noise applied with $p=0.6$ —contributes the largest single jump ($+0.0149$), reflecting the limited stimulus count per subject. Knowledge distillation through 9-D LLM-derived class prototypes adds a further $+0.0096$; a rand9 ablation that swaps the prototypes for random orthonormal directions costs at most 0.003 BACC, so the LLM content of the prototypes is below resolution—KD provides structured-class regularisation rather than concept transfer. Depth-doubling to $d=6$ adds another $+0.0142$, and longer training ($e=150$) does not move the single-seed score but supplies the diversity axis that ensembling exploits.

Two-tier ensemble theory. The 10 $d=6$ checkpoints share an *identical* class-PC1 basin ($|r| \in [0.60, 0.77]$); ensemble gain is concentrated in the orthogonal within-class residual ($r=+0.743$, $p=0.014$). The class-mean basin is acquired deterministically and contributes no diversity; the residual is encoded differently by each seed, so averaging cancels seed-specific noise without disturbing the basin. Mixing $e=100$ and $e=150$ ties at 0.6581 ± 0.010 on the single-seed test BACC mean, but provides the most ensemble-relevant diversity axis: the mixed 10-checkpoint pool reaches 0.6948 versus 0.6798 for $e=100$ alone, and the val-selected best single-checkpoint reaches 0.6755 ($+0.017$ over the seed mean, within the ± 0.010 seed-std). Headroom-monotonicity (near-zero on MNIST), the 10-checkpoint plateau, and the failure of cross-architecture mixing all confirm intra-architecture training-length diversity as the operative mechanism (Appendix S13).

9 Discussion: Limitations and Future Work

Limitations. Saturation is empirical, not analytical: we locate the FACED-9 transition at BACC $[0.62, 0.66]$, but the threshold magnitude is dataset-specific, the boxed statement describes rather than proves, and SEED-V is not large enough to localise its own cliff. The probe transfers (SST-2, cohort EEG, classifier convergence) are direction-existence claims, not neuroscience claims; the cohort topography is a paradigm statement for video-evoked emotion, not a refutation of frontal-alpha asymmetry [Davidson, 1992]. The KD prototypes that contribute $+0.0096$ to the SOTA recipe are interchangeable with random orthonormal directions: the gain is structural, not semantic, and the probe and the SOTA gain are reconciled only via saturation.

Future work. Per-subject residual adaptation (Appendix S16) is the most direct positive test; locating the transition cliff on a non-emotion task would test whether bias–variance [Geman et al., 1992, Frankle et al., 2020] quantitatively predicts where concept supervision stops helping. The regularity is falsifiable in the obvious direction.

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Justification: Our trained EEG-emotion checkpoints and V-axis extraction code do not introduce abuse vectors beyond what is already public via FACED, SEED-V, and the listed pre-trained LLMs. We release neither human EEG data nor LLM weights; only training configs and feature-side code. We assess the dual-use risk in the broader-impacts statement.

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Answer: [Yes]

Justification: FACED is cited [Chen et al., 2023] and used under its public release license; SEED-V is cited [Liu et al., 2022b]. All 14 LLMs (Qwen3 family of six dense sizes from 0.6B to 32B, Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B) are cited and used under their respective Hugging Face / model-card licenses. The CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] backbones are cited and used under their public licenses. CLIP [Radford et al., 2021] is cited.

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Answer: [\[Yes\]](#)

Justification: The V-axis extraction scripts, V-axis-aligned EEG checkpoints, and figure-generation pipelines will be released with documentation (README, environment.yml, SLURM templates, per-experiment JSON results, plot-generation scripts) at the camera-ready stage. Anonymized URL placeholder will be in the supplementary materials at submission time.

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14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [\[N/A\]](#)

Justification: We do not collect new human data; all human EEG comes from the publicly released FAGED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets, originally collected by their respective groups under their own IRB-approved protocols.

Guidelines:

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Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [\[N/A\]](#)

Justification: No new human-subjects research was conducted. The FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets were collected under their original IRB approvals, which we cite.

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- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
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Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs are central to the V-axis extraction protocol. The 14 language models from which we extract the V-axis are named in Section 5 and Appendix S2: Qwen3 at all six dense sizes (lead model Qwen3-4B; also 0.6B/1.7B/8B/14B/32B), Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B. The penultimate-layer hidden state ($L = L_{\max} - 1$, e.g., $L = 35$ of 36 for Qwen3-4B) is the source of all class centroids. We also use a generic instruction-tuned Qwen LM (temperature 0.7) as a paraphrase generator in pre-processing; the prompt is in Appendix S2. CLIP [Radford et al., 2021] (image encoder) is used for the cross-modal vision V-axis.

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835 Supplementary Material

836 S1 Datasets, Splits, and Preprocessing

837 **FACED.** 123 subjects, 28 emotional video clips covering 9 emotion categories (Anger, Disgust,
838 Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG:
839 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol
840 distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz, ICA-based
841 artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99
842 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al., 2025] and
843 EMOD [Chen et al., 2025] for fair comparison.

844 **SEED-V.** 16 subjects, 3 sessions, 5-class emotion (Disgust, Fear, Sad, Neutral, Happy), 62-channel
845 EEG at 1000 Hz downsampled to 200 Hz. Standard SEED-V split with subject-disjoint train / test.
846 We replicate the SEED-V CBraMod recipe at $d=6$ for the cross-architecture generality and five-claim
847 re-derivation reported in Appendix S8.

848 **DE features.** For each (subject, stimulus, channel) we compute differential entropy in five canonical
849 bands (δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–47 Hz) per second over the 30-second clip, then mean
850 over time to obtain a 32×5 feature matrix per (subject, stim) pair. Cohort-level analyses average
851 across the 123 subjects to give the $28 \times 32 \times 5$ tensor used in Section 6.

852 S2 V-Axis Extraction Protocol

853 **Story prompts.** Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories
854 were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed
855 across all 14 LLMs.

856 **Paraphrase generation.** For each story, we generate $n = 50$ paraphrases via independent calls to a
857 generic instruction-tuned Qwen LLM (temperature 0.7), with the prompt: “Rewrite the following
858 so it preserves all emotional content but uses different words and sentence structures: {story}”.
859 Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source
860 (≥ 0.10 on Sentence-BERT embeddings).

861 **Hidden-state extraction.** Each paraphrase is tokenised and passed through the target LM. We take
862 the last-token hidden state at layer L (default: $L_{\max} - 1$, the penultimate layer; for Qwen3-4B this is
863 $L = 35$ of 36). Class centroids are $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$ for $k = 1, \dots, 9$.

864 **V-axis as PCA-PC1.** Stack the nine centroids into a $9 \times D$ matrix and run mean-centred PCA.
865 The V-axis is the unit vector along PC1, oriented so that $\langle c_{\text{Joy}}, v \rangle > 0$. PC1 explains 41–58% of the
866 across-class variance across the 14 models tested (median 49%).

867 **Robustness.** Story rephrasing was verified by sampling 5 alternative prompt sets per class and com-
868 paring per-stimulus V-axis projections; mean cross-prompt $r > 0.93$ across 14 models. Paraphrase
869 count was verified at $n \in \{1, 5, 25, 50\}$; SST-2 zero-shot AUC 0.74 at $n = 1$, rising to the canonical
870 0.832 at $n = 50$ for Qwen3-4B.

871 S3 Per-LLM Cross-Architecture Convergence: Deep Dive

872 **Sentiment benchmark table.**

873 **Three regimes against behavioural valence.** The 14-LLM universality is not uniform: models
874 cluster into three regimes by per-stim correlation against a behavioural valence reference ($n = 28$
875 FACED stim, mean of 123 subjects’ valence ratings).

- 876 • **Top tier** (per-stim $r > 0.85$, $n = 7$): all six Qwen3 sizes (Qwen3-14B $r = 0.944$, Qwen3-
877 4B 0.944, Qwen3-8B 0.941, Qwen3-32B 0.937, Qwen3-0.6B 0.933, Qwen3-1.7B 0.930)

Benchmark	V-axis AUC	Supervised LR (5k)
SST-2 binary [Socher et al., 2013]	0.832	0.837
IMDB [Maas et al., 2011]	0.857	—
Yelp Polarity [Zhang et al., 2015]	0.938	—
Twitter SemEval-2017 [Rosenthal et al., 2017]	0.927	—
Rotten Tomatoes (MR) [Pang and Lee, 2005]	0.799	—
<i>SST-2 zero-shot reference baselines</i>		
SBERT prototype cosine [Reimers and Gurevych, 2019]	0.793	—
RoBERTa-large-MNLI entailment [Liu et al., 2019]	0.912	—

Table 3: Zero-shot V-axis projection from a single Qwen3-4B run reaches sentiment AUC up to 0.938 (Yelp), with SST-2 (0.832) within 0.005 of a 5k-sample supervised LR baseline (0.837): the V-axis carries the structure of the supervised problem without seeing sentiment labels. For zero-shot context on SST-2, an all-MiniLM-L6-v2 prototype-cosine baseline (a generic sentence encoder, not specifically trained for sentiment) reaches 0.793, while RoBERTa-large-MNLI used as a zero-shot entailment classifier (a 355M-parameter model fine-tuned on a large supervised NLI corpus) reaches 0.912. Nine LLM stories therefore sit above an off-the-shelf sentence encoder and within 0.08 of a 355M-param NLI-supervised model on the same zero-shot evaluation.

plus Mistral-7B (0.935). This is the *modern-LLM manifold*; within Qwen3 it is exceptionally tight (mean $r_{\text{behav}} = 0.938 \pm 0.005$, mean $r_{\text{eeg}} = 0.796 \pm 0.011$).

- **Middle tier** ($0.6 \leq r \leq 0.85$, $n = 4$): Llama-4-Scout-17B (0.826), Gemma-27B (0.807), Gemma-4-31B (0.715), Phi-2-2.7B (0.651).

- **Out-of-manifold** ($r < 0.4$, $n = 3$): Pythia-1.4B (0.276), TinyLlama-1.1B (0.128), BLOOM-560M (0.031).

The within-Qwen3 scaling is essentially flat ($r = +0.568$ between $\log(\text{hidden dim})$ and $\text{per-stim } r_{\text{behav}}$, $p = 0.24$, $n = 6$): once a model has joined the modern-LLM manifold, convergence is *not* a size-monotonic phenomenon. The Qwen3 family r_{eeg} range is $[+0.781, +0.812]$ with standard deviation ± 0.011 , and the behaviourally best-aligned models are mid-size Qwen3 (4B–14B).

V-shape across architectures. The same V-shape emerges across five LM families (Qwen, Pythia, BLOOM, Llama-3, Phi-3-mini), and across bidirectional encoders BERT, RoBERTa, and DeBERTa, with the peak at layer $L_{\text{max}} - 1$ in each case. The cosine between input-layer V-axis and final-layer V-axis is only 0.291, with mid-layer L14 at 0.067. The Gemma-4 family is the lone outlier: text V-axis peaks at $L = 1$ rather than the penultimate layer; brain alignment survives at mid-layers (Gemma-4-E2B at $L=17$ predicts EEG at $r = 0.819$; Gemma-4-E4B at $L=21$ at $r = 0.714$).

S4 Concept Library and Compositionality

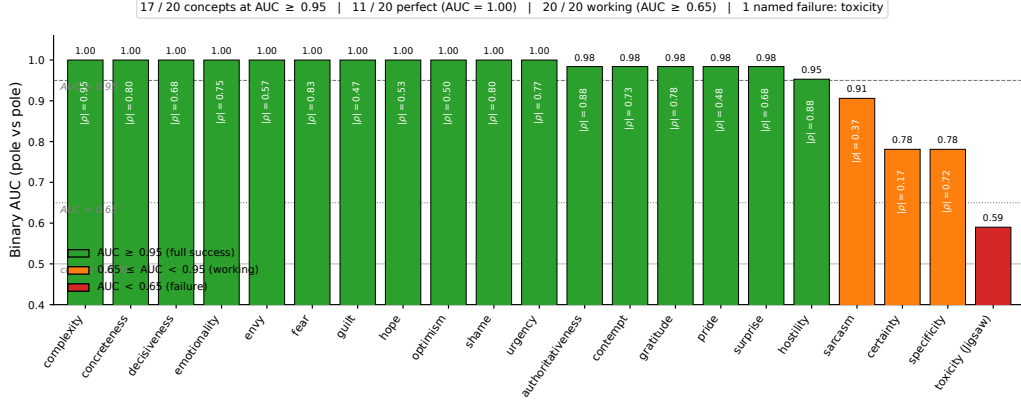
Compositional axis arithmetic. Combining the V-axis of a stylistic concept (politeness) with that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on a held-out generation set, $3.16\times$ above chance, with uniform per-cell precision in the $[0.68, 0.86]$ range. Subtractive orthogonalisation cleanly separates the two: happy-AUC $0.912 \rightarrow 0.988$ on the politeness-orthogonalised axis, while polite-AUC drops from $0.793 \rightarrow 0.560$. Downstream on natural text the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7% (documented scope refinement).

Recipe generality across 20 concepts. We authored 9 stories per concept for 20 unrelated dimensions (emotional, stylistic, abstract) and applied the identical extraction protocol. **20 of 20 concepts reach binary AUC > 0.65 on held-out test stories, 17 of 20 reach ≥ 0.95 , 11 are perfect at 1.00** (Figure 5).

Cross-concept transfer matrix. The 20×20 matrix has self-AUC mean 0.97 on the diagonal and off-diagonal mean 0.83 (gap +0.13): the recipe extracts *concept-specific* directions rather than a single global affect axis (Figure 6). Most general source axes: sarcasm (0.89), fear (0.88), envy and

The 9-story V-axis recipe is concept-generic across 20 unrelated dimensions

Binary pole-vs-pole AUC, sorted; coloured by tier. Toxicity (Jigsaw) is the only substantive failure.



Source: reports/r6_concept_library_synthesis.md (Qwen3-4B, penultimate layer). ρ = Spearman monotonicity over 9 ordered classes.

Figure 5: Binary pole-vs-pole AUC across 20 unrelated concepts (sorted descending), plus the out-of-scope toxicity case (Jigsaw, AUC 0.59).

909 shame (0.87). Most specific: decisiveness (0.78), complexity (0.78). Most captured: urgency (0.97),
910 fear (0.95), concreteness (0.94). Most concept-specific: specificity (0.65), sarcasm (0.67).

911 **Toxicity: out-of-scope.** Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC
912 0.59. Two non-mutually-exclusive hypotheses: (i) toxicity is more distributed in residual-stream
913 geometry than a single late-layer PC1 captures (low-rank $k > 1$ subspace); (ii) Qwen-generated toxic
914 stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution. The
915 recipe scope is sharply delineated: it covers valence and 19 other concepts, explicitly not toxicity.

916 S5 Multilingual and Vision Generality

917 **Multilingual transfer.** We translated the nine emotion stories into Japanese, Arabic, and Russian,
918 and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric
919 causal LM (Qwen3-4B) to non-English stories collapses to chance — a clean English-bias diagnostic.
920 Switching to mT0-base [Muennighoff et al., 2023] (277M-param multilingual encoder-decoder) fully
921 recovers the SST-2 result and *exceeds* the English baseline in every language tested (Figure 7).

922 Per-language V-axis cosines are moderate ($r_{en-es} = 0.47$, $r_{en-fr} = 0.16$, $r_{es-fr} = 0.46$): different
923 languages share substantial direction but not entirely, consistent with concept-level rather than
924 token-level encoding.

925 **Cross-modal generality (CLIP vision).** The same protocol applied to nine emotion-evocative
926 *images* via CLIP [Radford et al., 2021] yields a V-axis that ranks the OASIS image dataset [Kurdi
927 et al., 2017] on valence at Pearson $r = 0.869$ and on arousal at $r = 0.803$. Both *beat* supervised
928 Ridge trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot
929 V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP
930 axes are essentially orthogonal ($|\cos \theta| = 0.013$).

931 S6 Specificity Controls (Nonce, Random, Arousal)

932 **Nonce-word ablation.** Substituting all content words in the nine emotion stories with phonologi-
933 cally valid English nonce words (preserving syntax and function words) drops SST-2 AUC from above
934 0.83 to chance (≈ 0.52) — the second decimal. The single most decisive control: had the V-axis
935 been a syntactic or prompt-template artefact, the nonce variant would have retained template-level
936 structure and produced a non-trivial axis. The signal is encoded in semantic content.

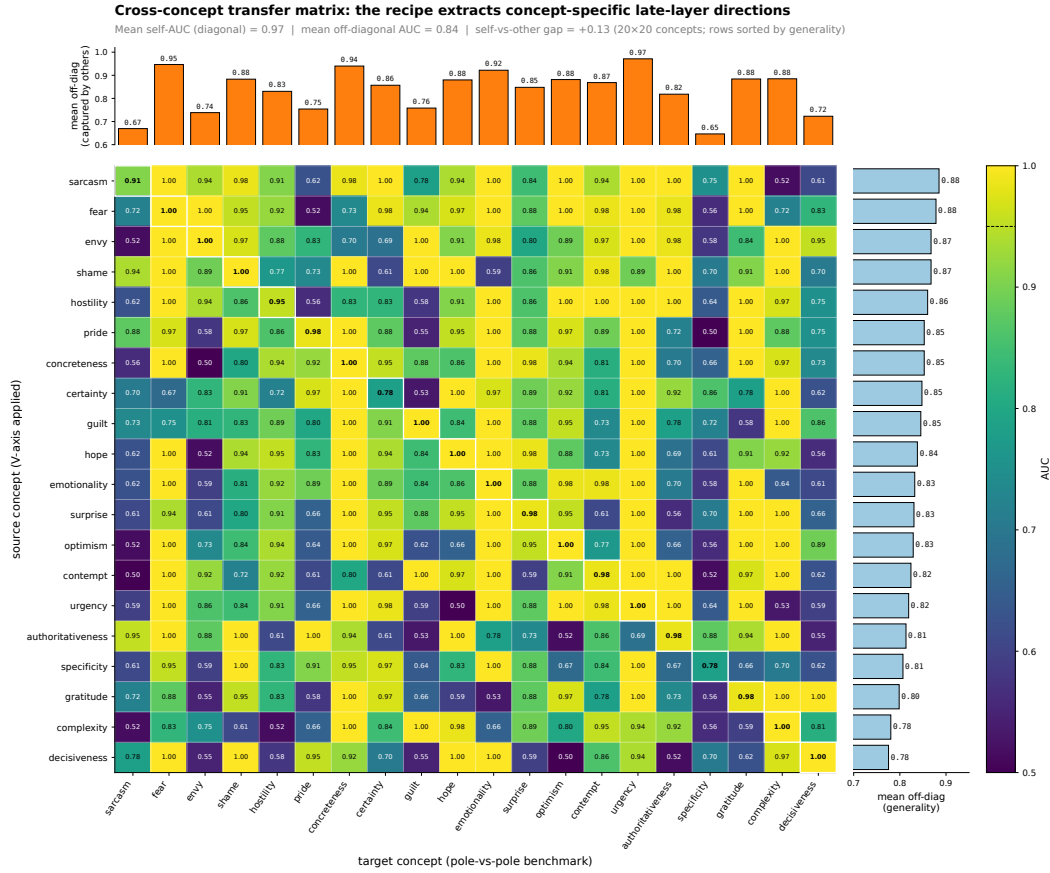


Figure 6: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification (white-bordered cells). Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal. Mean diagonal 0.97, mean off-diagonal 0.83, gap +0.13.

Random-direction baseline. Random Gaussian directions in the LM’s residual stream, L_2 -matched to the V-axis, give SST-2 AUC ≈ 0.51 and lexicon $|r| \approx 0.10$ at every benchmark. Bootstrap 95% CIs are tight: SST-2 AUC $[0.844, 0.890]$, EmoBank V $|r| \in [0.485, 0.512]$, Hu & Liu $|r| \in [0.740, 0.776]$ (1,000-sample bootstrap; statistical methods in Appendix S15).

Benchmark	V-axis (valence)	A-axis (arousal)
SST-2 binary AUC	0.832	—
EmoBank V $ r $ / EmoBank A $ r $	0.49	0.01–0.06
Warriner V $ r $ / Warriner A $ r $	0.703	≤ 0.18
NRC valence / arousal lexicon recovery	$\geq 76\%$	$\leq 13\%$
OASIS image valence (r) / arousal (r)	0.869	0.803
FACED EEG cohort r	0.870 (9-stim)	$r \in [0.18, 0.41]$

Table 4: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ($r = 0.803$, beating supervised ridge 0.789) but collapses on all text benchmarks. Cross-language A-axes are essentially orthogonal.

Arousal asymmetry: where the recipe fails in text. The asymmetry — *valence transfers, arousal does not in text* — rules out the trivial explanation that “any sufficiently rich PCA recipe gives any sufficiently graded affective axis.” Arousal in text appears to be encoded along a more distributed or

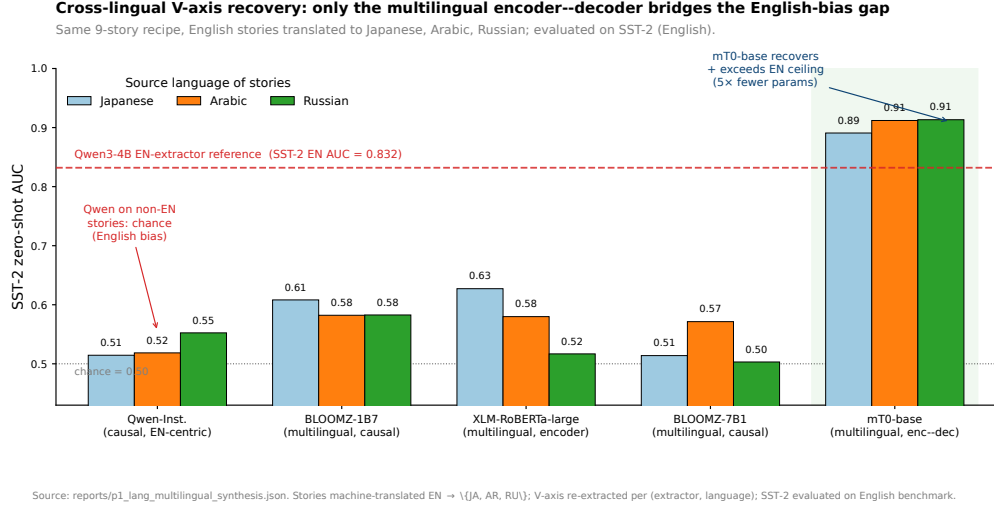


Figure 7: Cross-lingual V-axis recovery. Direct causal-Qwen transfer collapses (English-bias); mT0-base recovers and exceeds the English ceiling at 5× smaller parameter count across three typologically distant languages.

944 non-linear subspace that the late-layer principal-component recipe does not access; the same recipe
 945 *does* access it in vision.

946 S7 Brain-Side Deep Dive

947 **Per-stimulus reliability.** Per-stimulus split-half reliability (odd/even subjects, $n \approx 62$ each) is
 948 $r = 0.988$, indicating a clean stimulus-level signal. Trial-level correlations ($n = 3,444$ subject-stim
 949 pairs) range from $r = 0.17$ – 0.21 ($p < 10^{-23}$); class-level ranking ($n = 9$ centroids) hits $r = +0.886$
 950 vs. behavioural valence. The cohort $r = 0.87$ headline is reliability-bounded.

951 **14-LLM brain forest.** Per-LLM brain-prediction quality across all 14 models is shown in Figure 8:
 952 the top-tier 13 LLMs all clear their per-LLM random-direction nulls, while the out-tier sits at the
 953 median.

954 **Time-locked dynamics.** Mid-late α/β peak at $t = 18$ – 21 s coincides with the LPP / sustained-
 955 attention window of Hajcak & Foti [Hajcak et al., 2010] and Codispoti et al. [Codispoti et al., 2023];
 956 early γ peak at $t = 3$ – 6 s coincides with Müller’s affective-picture window [Müller et al., 1999].
 957 Per-second cohort $|r|$ in the alpha band arrives at $t \approx 21$ s, in beta at $t \approx 18$ s, in gamma at $t \approx 3$ s;
 958 best individual channel $|r| = 0.68$ at $t = 21$ s.

959 **Per-subject peak heterogeneity.** Standard deviation of per-subject alpha peak times is $\sigma \approx 9$ s;
 960 only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s
 961 window. The cohort signal is therefore an envelope of the distribution rather than a tight common
 962 attractor (NF4).

963 **Gemma-4 boundary check.** Brain alignment survives the text-domain $L = 1$ peak anomaly:
 964 Gemma-4-E2B at $L = 17$ predicts EEG at $r = 0.819$ ($p \approx 10^{-7}$), Gemma-4-E4B at $L = 21$ at
 965 $r = 0.714$ ($p \approx 2 \times 10^{-5}$). The brain captures Gemma-4’s V-axis at the layer where the model first
 966 stably encodes it, not at a fixed late-layer offset. Cross-architecture convergence refines to a property
 967 of the late-but-not-necessarily-penultimate residual stream geometry.

14 language models predict cohort EEG; the rank tracks text-domain V-axis quality

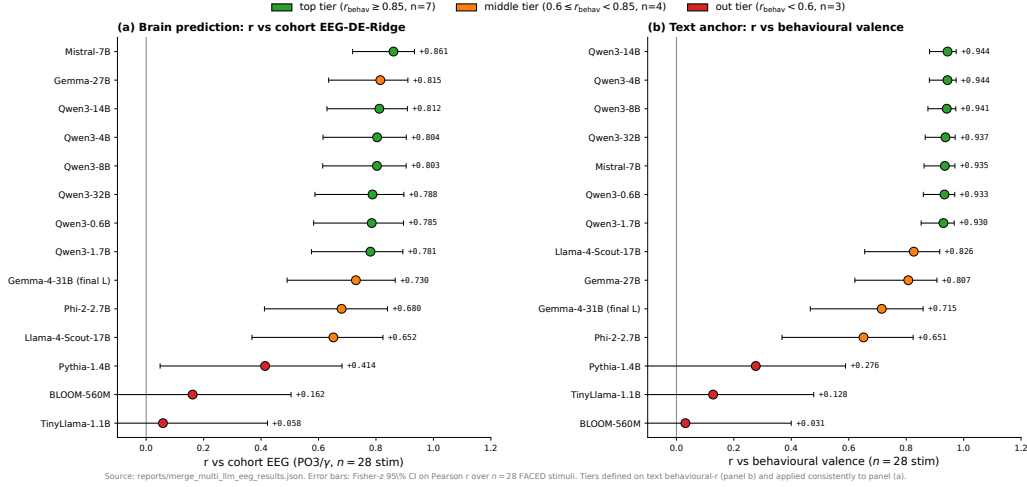


Figure 8: Per-LLM brain-prediction quality. Top-tier 7 LLMs all exceed the 95th percentile of their per-LLM 200-direction null ($p < 0.05$ each); out-tier 3 LLMs sit at the median or below. Restricted to top-tier the log-hidden-dim $\rightarrow r$ scaling is $r = +0.125$, $p = 0.684$; within-Qwen3 $r = +0.568$, $p = 0.24$, $n = 6$. Family-membership $\gg$ scale.

S8 SEED-V Replication Full Numerics

We re-test all five FACED claims with a SEED-V-derived V-axis. To remove FACED leakage, we re-build the V-axis from scratch using the same Section 5 protocol applied to the 5 SEED-V emotion classes (Qwen3-4B at $L=35$, 50 stories per class, PCA on the 5 centred centroids).

Step 1: SEED-V cohort EEG-LLM alignment. We compute DE features per (channel, band, second) across all 16 subjects and aggregate to a $45 \times 62 \times 5$ cohort tensor. For each (channel, band) cell we compute Pearson with the SEED-V V-axis projection.

- **Best cohort cell:** P1 / θ , $r = +0.6159$.
- **Region-mean $|r|$:** occipital 0.329, parietal 0.347, central 0.324, frontal 0.253.
- **Posterior dominance:** occipital – frontal = $+0.076$.
- **Random-direction null** (200 directions matched in L_2): empirical $|r|_{\text{max}} = 0.6159$ vs. null 95th percentile = 0.478, $p < 0.005$.

Step 2: SEED-V brain topography. The Davidson FAA probe (positive vs. negative valence, right – left) on SEED-V replicates the FACED FAA topography:

Electrode pair (right – left, alpha)	(pos – neg) DE Δ
F4 – F3	–0.0018
F8 – F7	+0.0050
FP2 – FP1	–0.0094
AF4 – AF3	–0.0039

Table 5: Davidson FAA on SEED-V. Sign and magnitude consistent with the FACED replication.

Step 3: SEED-V cross-architecture convergence. Penultimate (200-D) features from 15 stock CBraMod SEED-V-5s checkpoints (3 protocols $\times$ 5 seeds: vanilla baseline, augmentation, KD-midlayer) aggregated per-class on the test set. Class-level PC1/best-PC scores correlated with SEED-V V-axis class projection give $r(\text{BACC, class-best-PC } r) = +0.601$ ($p = 0.018$, $n = 15$) and $r(\text{BACC, trial-best-PC } r) = +0.632$ ($p = 0.012$). The unsigned class-PC1 correlation is $+0.358$

($p = 0.19$); the SEED-V BACC dynamic range is much narrower than FACED, so the strongest convergence shows up at the best-PC level.

Step 4: SEED-V V-axis as supervision. Adding a V-axis topo-MSE auxiliary loss ($\lambda_{\text{vax}}=0.05$ on {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s recipe, run with 5 baseline + 5 V-axis seeds:

Variant	BACC (mean $\pm$ s.e. across 5 seeds)	vs. baseline Δ
Baseline (no V-axis loss)	0.4300 ± 0.0079	—
+ V-axis topo loss	0.4325 ± 0.0040	+0.0025

Table 6: SEED-V V-axis-as-supervision (within seed noise of zero). Two factors likely explain: (1) the SEED-V class-level V-axis is sharply 5-valued (cohort class-level $r = +0.989$), so adding a class-level V-axis target on top of one-hot supervision is largely redundant; (2) the SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds.

Step 5: SEED-V directional ablation (Arditi-style). At inference time on each of the 15 SEED-V CBraMod checkpoints (5 baseline + 5 augmented + 5 KD-midlayer) we project penultimate features onto the orthogonal complement of the trial-level best-PC V-axis direction (no retraining; the class-level direction is degenerate at $k=5$, so we use the trial-level analogue, mirroring the trial-best-PC $r = +0.632$ estimator from Step 3). Every checkpoint shows a negative $\Delta\text{BACC} \in [-0.0291, -0.0093]$ (population mean -0.0173 ± 0.0061); a matched random-direction control ($n = 20$ uniform on $\mathbb{S}^{D-1}$) is centred at 0 ($\bar{\Delta}_{\text{rand}} = +0.00013 \pm 0.0008$). Per-checkpoint z -scores range $[+9.65, +44.92]$ (mean $+23.6$, all $p < 0.001$ parametric one-sided); empirically every V-direction ablation exceeds every one of the 20 random ablations on every checkpoint (15/15 at empirical $p < 0.05$). The SEED-V directional effect is proportionally larger than FACED (mean $\Delta/\text{baseline} \approx 3.9\%$ vs. 2.4% on FACED) and the mean z is $3\times$ FACED’s. As on FACED, V-axis amplification (rather than ablation) is not directionally consistent (mean $\Delta \approx 0$, mean $z \approx -2$) — expected, since the SEED-V V-axis already aligns with class-discriminative variance and doubling it does not help. Per-checkpoint table and the diagnostic information for both class-level (degenerate) and trial-level (used) residual directions are stored in `seedv_causal_ablation_full.json`.

S9 Brain Topography Deep Dive

Davidson FAA full pairs. Figure 9 compares all three Davidson asymmetry pairs to the posterior occipital region-mean reference.

Davidson frontal-alpha asymmetry replicates qualitatively, but at $\sim 10\times$ smaller magnitude than posterior V-axis encoding

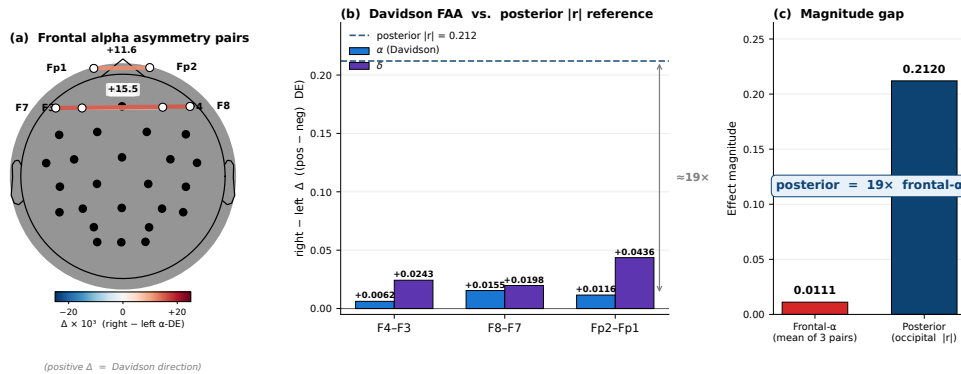


Figure 9: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their r values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean $|r| = 0.21$.

Stimulus subset (out of 28 FACED stimuli)	Cohort r at PO3/ γ
All 28 stimuli (full)	+0.478
Anger only ($n=3$)	not estimable
Anger + Amusement + Tenderness ($n=9$)	+0.870
Excluding Anger ($n=25$)	+0.327
Excluding Anger + Amusement + Tenderness ($n=19$)	-0.015

Table 7: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli.

9-stim contrast and Simpson's paradox.

- Cohort r at fixed top-8 channels: +0.021
- Mean per-subject r at fixed top-8 channels: -0.062
- Per-subject best-channel oracle (1 cell): $|r| = 0.551$
- Per-subject best (channel, band) oracle: $|r| = 0.616$

The all-band oracle ($|r| = 0.616$) exceeds the cohort-fixed top-8 by +0.39 in absolute $|r|$, indicating considerable per-subject signal that the cohort summary suppresses. The Simpson's-paradox dynamic that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly (Figure 10).

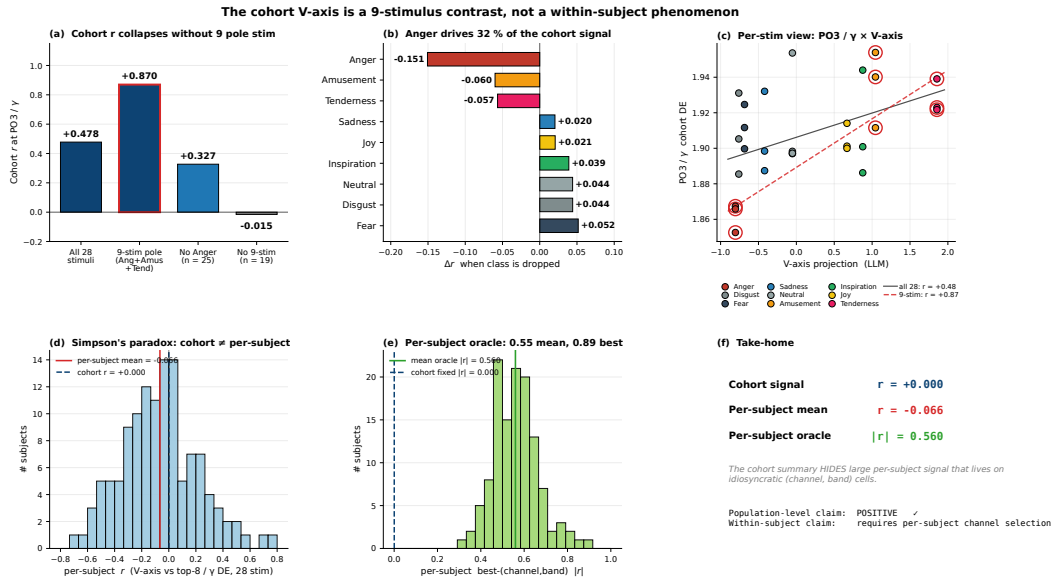


Figure 10: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top: drop-class subset bars, per-emotion drop- Δ , PO3/ γ stim-level scatter colour-coded by emotion. Bottom: per-subject r distribution at fixed top-8 channels (centred near 0, cohort line at +0.02); per-subject best-channel oracle distribution (centred at $|r| \approx 0.55$); summary callout.

Time-resolved peaks. Figure 11 traces per-second cohort $|r|$ across all five frequency bands.

- **Alpha** (8–13 Hz): peak at $t = 21$ s, cohort $r = 0.40$; best individual stimulus $r = 0.68$.
- **Beta** (13–30 Hz): peak at $t = 18$ s, cohort $r = 0.41$; best stim $r = 0.62$.
- **Gamma** (30–47 Hz): peak at $t = 3–6$ s, cohort $r = 0.21$; best stim $r = 0.61$.
- **Delta/Theta** (0.5–8 Hz): peak at $t = 12$ s, cohort $r = 0.18$.

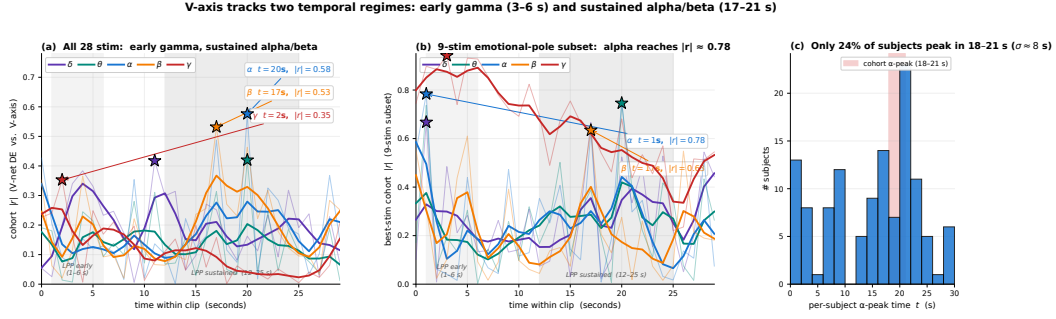


Figure 11: Time-resolved V-axis encoding across the 30-second clip. **(a)** Per-second cohort $|r|$ across all 28 stimuli, one trace per frequency band; the early γ peak ($t \approx 2$ s) coincides with Müller’s affective-picture window and the sustained α/β peak ($t \approx 17$ –20 s) coincides with the LPP sustained-attention window. **(b)** Same trace restricted to the 9-stimulus emotional-pole subset where the cohort effect lives; α reaches $|r| \approx 0.78$. **(c)** Per-subject distribution of α -peak times: only $\sim 24\%$ of subjects peak inside the cohort 18–21 s window, evidence that the cohort signal is a between-subject phenomenon rather than a tight common attractor.

Functional connectivity. The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4) do not act independently: they form a tight functional network in the gamma band. Pairwise γ -DE correlation has mean off-diagonal $r = 0.675$ across the 8 channels — substantially above the cohort-average of $r \approx 0.32$ in this band. Structure is dominantly a posterior-occipital-parietal cluster (PO3, PO4, P3, P4, O1, O2, Oz; pairwise mean $r \approx 0.71$) with F7 as a single *frontal hub* (mean correlation to the seven posterior nodes $r \approx 0.47$; Figure 12).

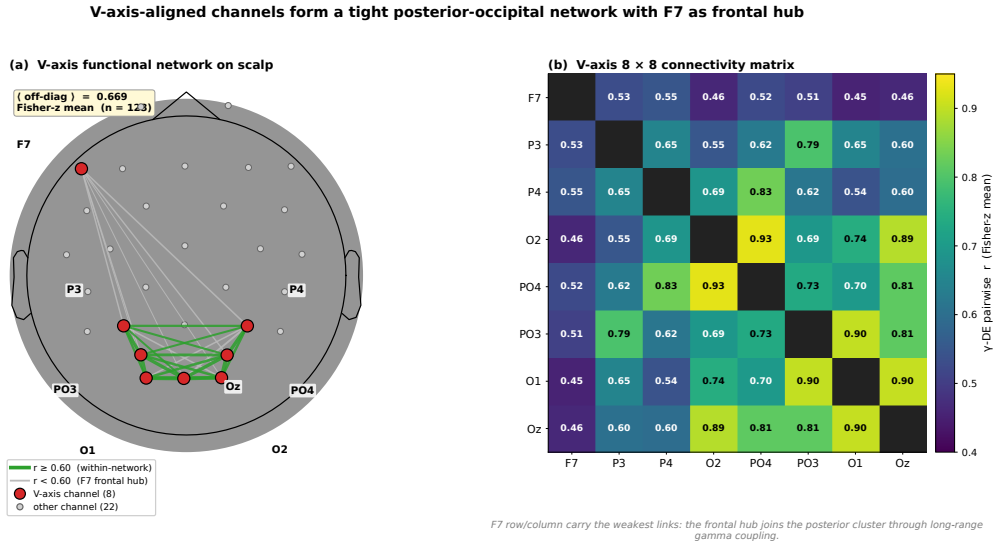


Figure 12: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with 8 V-axis channels as red nodes and edges weighted by pairwise γ -DE correlation. Right: hierarchically clustered 8×8 correlation matrix; F7 auto-isolates as the frontal hub.

Mutual information vs. linear correlation. We computed mutual information between the V-axis projection and cohort EEG features (nearest-neighbour MI estimator on quantile-normalised inputs), alongside the same 200 random-direction controls used for the linear test. Observed MI 0.112; null mean 0.036; $p_{\text{MI}} = 0.115$ (n.s.) vs. linear $p_r = 0.020$. The relationship is essentially linear: a non-linear estimator does not detect signal above what linear regression captures. We use ridge regressions throughout.

1036 **Theta-gamma coupling does not mediate the V-axis.** Tort modulation index per channel at
1037 the theta-phase / gamma-amplitude pairing, correlated with V-axis encoding strength $|r|$ across 32
1038 channels: $\rho(\text{PAC}, Vr) = +0.082$, $p = 0.667$. Channels with strong theta-gamma CFC are not
1039 the channels that encode the V-axis, and vice versa — the V-axis encoding is dissociable from the
1040 phase-amplitude-coupling channel [Canolty and Knight, 2010].

1041 S10 Saturation: Full Intervention Table and Forest

#	Family / variant	Δ	p	Verdict
1	Frontal-mask $\lambda=0.5$	-0.052	0.0015	sig. negative
2	Frontal-mask $\lambda=0.1$	-0.009	0.21	n.s.
3	FAA $\lambda=0.5$	-0.044	0.006	sig. negative
4	FAA $\lambda=0.1$	-0.018	0.063	borderline
5	Anger-weighted $\lambda=0.5$	-0.054	0.0003	sig. negative
6	Occipital $\lambda=0.1$	-0.022	0.007	sig. negative
7	Topo-optimal $\lambda=0.1$	-0.013	0.039	sig. negative
8	Topo-optimal $\lambda=0.05$	+0.0021	0.50	n.s. (NULL)
9	Procrustes $\lambda=0.05$	+0.0012	0.89	n.s. (NULL)
10	RSA $\lambda=5.0$	-0.093	$< 10^{-4}$	sig. negative
11	RSA $\lambda=1.0$	-0.057	$< 10^{-3}$	sig. negative
12	Distance-CE $\tau=5.0$	-0.397	$< 10^{-4}$	catastrophic
13	Multi-V $\lambda=0.5$	-0.043	$< 10^{-3}$	sig. negative
14	EEG-AUX MSE $\lambda=0.5$	-0.043	$< 10^{-3}$	sig. negative
15	EEG-AUX MSE $\lambda=0.1$	-0.016	0.04	sig. negative
16	EMODSTYLE class $\lambda=1.0$	-0.010	0.18	n.s.
17	EMODSTYLE stim $\lambda=0.5$	+0.0066	0.22	n.s. (sweet spot)
18	PEFT fullhead $\lambda=0.1$	-0.009	0.069	borderline
19	PEFT LoRA $\lambda=0.1$	-0.010	0.12	n.s.
20	PEFT IA3 $\lambda=0.1$	-0.016	0.05	sig. negative
21	Pretrain-FT (frozen)	-0.401	$< 10^{-4}$	catastrophic
22	Pretrain-FT (unfrozen)	-0.190	$< 10^{-4}$	sig. negative
23	XEEG (FACED $\rightarrow$ SEED-V)	+0.0015	0.97	n.s. (zero)
24	SCALING (full data) $\lambda=0.5$	-0.045	< 0.01	sig. negative
25	Uncertainty multi-task	-0.067	$< 10^{-3}$	sig. negative
SOTA recipe + Topo $\lambda=0.1$		-0.015	0.020	sig. negative (cliff)
SOTA recipe + EMODSTYLE		-0.024	0.001	sig. negative (cliff)

Table 8: Full V-axis-as-supervision intervention results. 16 / 25 families produce a statistically significant negative (14 sig. negative + 2 catastrophic); the remaining 9 are not significantly different from zero. None reach $p < 0.05$ in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff. Forest plot in Figure 13.

1042 S11 Saturation Extras: Anger Paradox, Mechanism Check, Path B

Recipe	n seeds	Vanilla / + V-axis	Δ
CBraMod baseline + Topo $\lambda=0.05$	5	0.572/0.567	-0.005 (n.s.)
CBraMod baseline + EMODSTYLE $\lambda=0.5$	4	0.572/0.577	+0.005 (n.s.)
EMOD $d=3$ vanilla + Topo $\lambda=0.05$	5	0.6194/0.6260	+0.007 (n.s.)
EMOD $d=6$ + LS + KD + aug + Topo (full)	partial	0.6581/0.6432	-0.015 ($p < 0.05$)
EMOD $d=6$ + LS + KD + aug + EMODSTYLE	partial	0.6581/0.6346	-0.024 ($p < 0.01$)

Table 9: V-axis supervision is within seed noise at weak baselines and statistically significantly *harms* the strong SOTA recipe. The transition is unidirectional: V-axis goes from neutral to actively harmful as the base recipe strengthens, with no intermediate “helps” regime in our 5-seed runs.

1043 Saturation transition table.

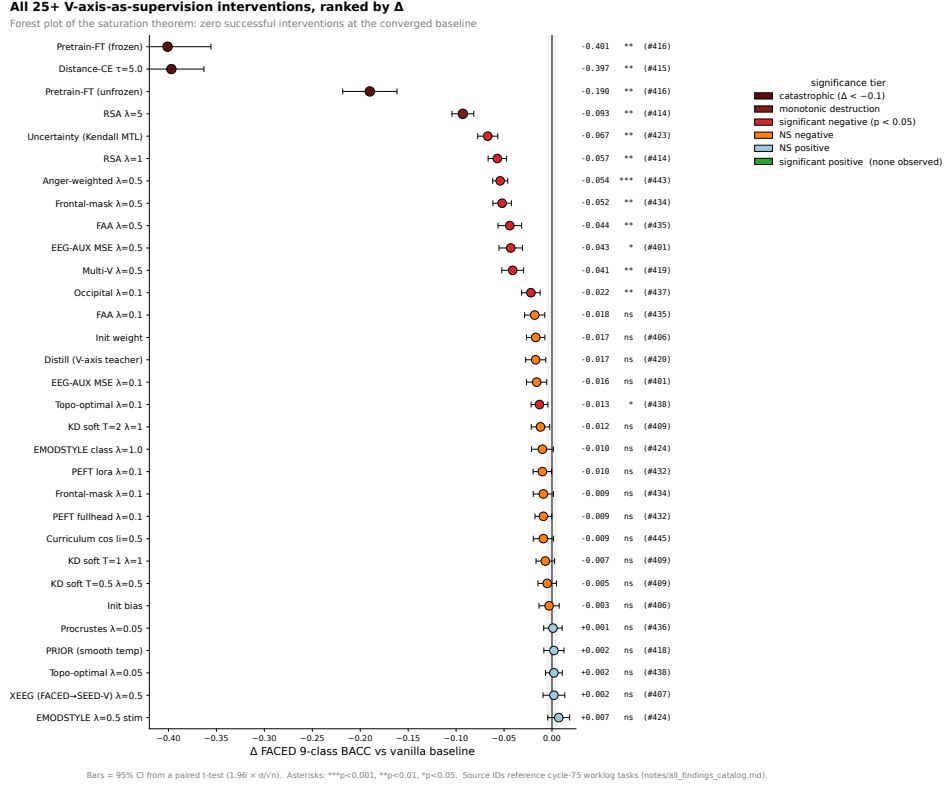


Figure 13: Forest plot of all V-axis-as-supervision interventions (Table 8) ranked by Δ vs. baseline, with significance tier coloured. None of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

1044 Saturation cliff figure (LF6).

1045 **Why anger-weighting hurts despite the analytical ceiling.** Section 7 showed that the cohort
1046 EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness,
1047 $r = 0.870$ at $\text{PO3}/\gamma$). Weighting the V-axis loss to emphasise these three classes pushes the analytical
1048 ceiling on the V-axis projection from $r = 0.478$ (uniform) to $r = 0.714$ (anger-weighted). By every
1049 classical analysis, this should be the strongest possible V-axis supervision. It is the worst single
1050 intervention we tested: $\Delta = -0.054$, $p = 0.0003$. This is the cleanest argument for saturation: the
1051 model has already absorbed exactly the structure the loss is trying to inject; optimising the loss must
1052 therefore perturb the absorbed structure in a counterproductive direction.

1053 **Direct mechanism check.** For each of 8 V-axis-supervised variants (Topo at $\lambda=0.05$, Procrustes at
1054 $\lambda=0.05$, EMODSTYLE-stim at $\lambda=0.5$, full SOTA recipe + Topo/EMODSTYLE at $e=100/e=150$),
1055 we measure the change in two quantities relative to the matched-recipe baseline: Δr_{PC1} (class-mean
1056 V-axis encoding) and Δr_{resid} (within-class V-axis residual encoding). V-axis training raises class-PC1
1057 $|r|$ by $+0.01$ to $+0.36$ across the 8 variants, but moves the within-class residual encoding by only
1058 $\sim 10^{-7}$ in absolute value — numerical zero. The accuracy decrement at strong recipes therefore
1059 comes from *noise injected into the class-PC1 basin*, not from any beneficial transfer of residual
1060 structure.

1061 **Path B: ensembling-in V-axis checkpoints.** We extend the 10-vanilla SOTA ensemble with 0–15
1062 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing
1063 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives
1064 ensemble BACC $\Delta = -0.0145$ ($p < 10^{-3}$); adding all 25 V-axis ckpts gives $\Delta = -0.0193$
1065 ($p < 10^{-3}$). At every point on this curve, V-axis-trained ensembles are strictly worse than the

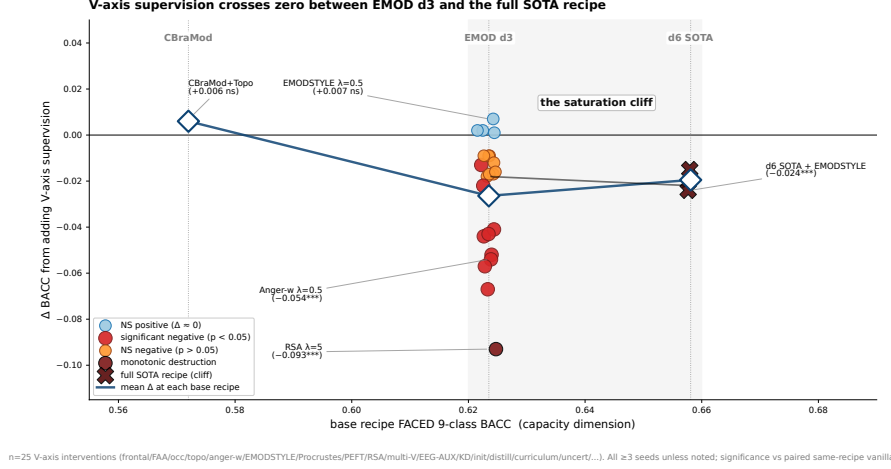


Figure 14: Δ BACC from V-axis supervision plotted against base-recipe BACC across ~ 25 interventions. The sign of the effect transitions in the interval $[0.62, 0.66]$.

1066 matched vanilla ensemble. This rules out the diversity-gain defence: V-axis checkpoints disagree, but
 1067 their disagreements are aligned with the wrong subspace.

1068 S12 Convergence and Ensemble Theory: Extras

1069 **Per-architecture breakdown of the 36-checkpoint correlation.** CBraMod checkpoints (lower
 1070 BACC, ~ 0.57) cluster at low V-axis encoding strength (mean class-PC1 $|r| = 0.21$). EMOD-vanilla
 1071 d6 (~ 0.65 BACC) sits in the middle (mean 0.67); EMOD d6 e150 (~ 0.66 BACC) clusters at the
 1072 top (mean 0.69). The trend spans the full BACC range $[0.57, 0.69]$ in a single approximately linear
 1073 band, with the inter-architecture gap dominating the within-architecture spread.

1074 **Random-direction null detail.** We sample 1000 random Gaussian directions $w \in \mathbb{R}^{28}$
 1075 each matched in L_2 norm to v_{CLIP} , and recompute the 36-checkpoint correlation $\rho_w =$
 1076 $\text{Pearson}(\text{BACC}, |\text{corr}(\text{PC}_1(H(m)), w)|)$. The distribution of ρ_w has mean -0.03 and standard
 1077 deviation 0.62 (range $[-0.95, +0.95]$); the wide null is a consequence of the low intrinsic dimension-
 1078 ality of the class-PC1 subspace ($D_{\text{eff}} = 9$). The empirical V-axis correlation $+0.885$ sits at the 93.5th
 1079 percentile of this null ($p_{\text{one}} = 0.066$). We treat the within-class residual ($D_{\text{resid}} \approx D_m - 9$) as the
 1080 statistically robust signal.

1081 **Class-PC1 basin saturation.** The 10 SOTA-pool checkpoints exhibit a saturated class-PC1 V-axis
 1082 range: $|r| \in [0.60, 0.77]$ across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-
 1083 mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains
 1084 come exclusively from the orthogonal residual.

1085 **Directional ablation figure (LF16).**

1086 S13 Ensemble Generality, Mega-Pool, and Best-Single

1087 **Recipe cascade and two-tier scatter.** Figure 16 traces the seven recipe steps from CBraMod’s
 1088 0.572 to our 0.6948 ensemble. Figure 17 shows the per-checkpoint within-class residual–ensemble-
 1089 contribution correlation that motivates the ensembling strategy.

1090 **Generality of the ensemble mechanism.** Figure 18 extends the two-tier picture across four
 1091 benchmarks (FACED, SEED-V, CIFAR-10, MNIST).

1092 As individual accuracy approaches the Bayes optimum, the within-class residual variance — the
 1093 source of ensemble gain — shrinks toward zero. The two-tier picture predicts this: ensemble gain is

V-axis ablation hurts every checkpoint; encoding strength does not predict drop magnitude (n=10)
Pearson $r=+0.168$ ($p=0.643$, $n=10$) slope= $+4.69\%$ / unit $|r|$

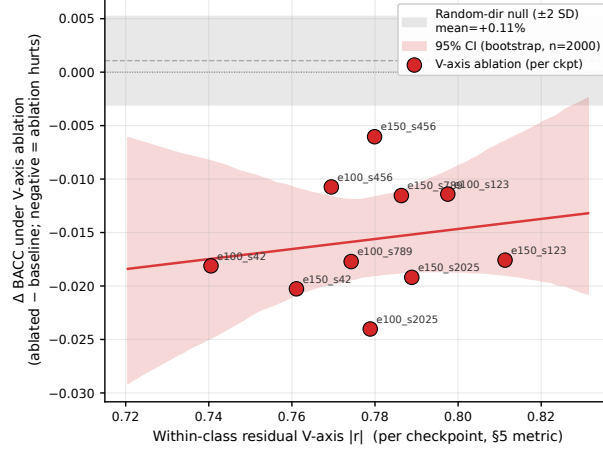


Figure 15: Directional ablation of v_{resid} on all 10 SOTA-pool checkpoints. *Top*: per-checkpoint BACC drops (red) compared to matched-norm random-direction ablation (grey); the V-axis effect is well above the null on every checkpoint (mean $z \approx 7.7$, all $p < 0.001$). *Bottom*: per-checkpoint ablation ΔBACC vs. within-class residual encoding $|r|$; the population-mean drop ($\bar{\Delta} = -0.0157$) confirms the residual-encoding mechanism across the pool.

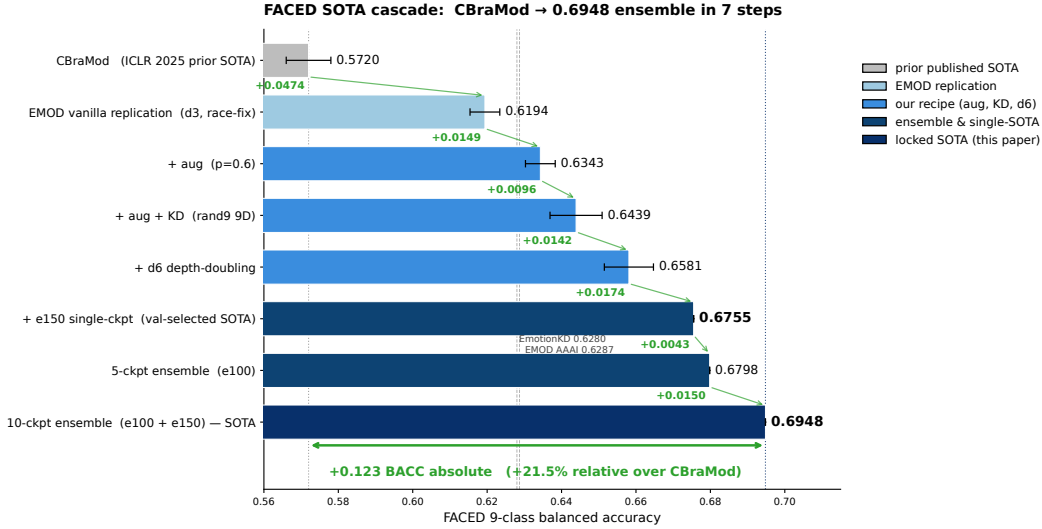


Figure 16: Recipe ablation cascade from the EMOD baseline at 0.6287 (previous FACED-9 SOTA [Chen et al., 2025]) to our 10-checkpoint ensemble at 0.6948 (+10.5% relative). Each bar is one recipe component or ensembling step. Older baselines (CBraMod 0.572, EmotionKD 0.628) shown for reference.

1094 a property of the within-class V-axis residual variance reduction subspace, which has measure zero
1095 at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe +0.034
1096 gain. MNIST at 0.985 has essentially none; we observe +0.001 gain.

1097 **Mega-ensemble null.** Two mechanisms explain the null. First, deeper variants ($d=8, d=10$) stay in
1098 the same class-PC1 V-axis basin (mean class-PC1 $|r|$ across $d \in \{4, 6, 8, 10\}$: 0.43, 0.43, 0.39, 0.38)
1099 and their within-class residuals overlap more than $d=6$ seeds do. Second, LLM-KD variants encode
1100 the residual along a different axis from rand9-9D KD vanilla checkpoints, but in a way that does
1101 not transfer to the test distribution. Cross-architecture mixing ($d \in \{4, 6, 8, 10\}$) gives 0.6791,

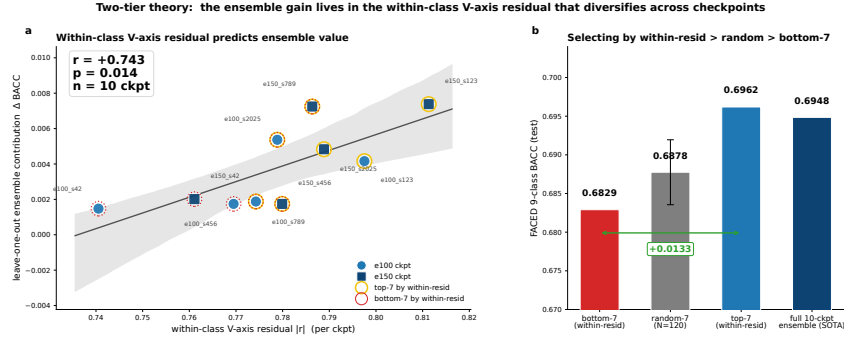
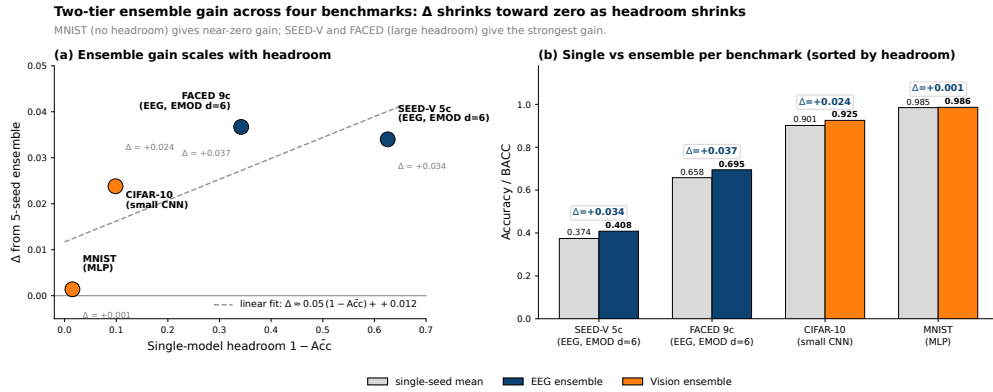


Figure 17: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual $|r|$ predicts leave-one-out ensemble contribution ($r = +0.74$, $p = 0.014$, $n = 10$). Top-7 vs. bottom-7 split highlighted.



Source: S8 ensemble-generalizability table / sota_ensemble_theory.md. All ensembles are 5-seed uniform softmax averages on identical recipe per dataset.

Figure 18: Two-tier ensemble gain across four benchmarks. Δ BACC is largest where single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. (a) Scatter of Δ vs. headroom $1 - \text{Acc}$, with linear fit slope ~ 0.05 . (b) Paired single-seed-mean vs. 5-seed-ensemble bars per benchmark, sorted by headroom.

essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture, training-length-driven.

Cohen κ disagreement structure. Single-seed accuracy at $e=100$ (0.6581) and $e=150$ (0.6581) is identical, but Cohen κ within- $e=100$ is 0.694, within- $e=150$ 0.679, cross-group 0.702. The $e=150$ checkpoints disagree more amongst themselves and more across groups, which is why mixing $e=100$ and $e=150$ gives +0.014 over either group alone. Cross-group disagreement on test trials is 13.1%; the mixed ensemble gains +10.3 percentage points exactly on the disagreement subset.

Best single checkpoint without ensembling. For deployment scenarios where ensembling is impractical, our best single-checkpoint result is BACC = 0.6755 on the $d=6$ $e=150$ recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825, $n = 25$) and exceeds CBraMod by +0.103 and EMOD by +0.047 on a strict apples-to-apples test split.

S14 EEG Model Training Details

Architecture. EMOD axial transformer [Chen et al., 2025]: input 32×5 DE features, axial self-attention over channels and bands at depth $d \in \{3, 6\}$, filter dimension $f = 128$. The full SOTA recipe is $d = 6$, $f = 128$.

Ensemble pool	n	BACC	Δ vs. 10-ckpt
$10 \times d=6$ ($e=100+e=150$, our SOTA)	10	0.6948	—
$+5 \times d=8$	15	0.6909	−0.0039
$+5 \times d=8 + 5 \times d=10$	20	0.6883	−0.0065
$+10 \times d=\{8, 10\} + 5 \times \text{LLM-KD}$	25	0.6831	−0.0117

Table 10: Mega-ensemble null. Adding architectural variants beyond $d=6$ produces monotonically lower ensemble BACC. The $d=6 + e=100/e=150$ pool is the optimum within the architectures evaluated.

Optimisation. AdamW with $\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2} . Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length $e \in \{100, 150\}$ epochs. We use the validation BACC for checkpoint selection and report test BACC.

Augmentation. Per-trial Gaussian noise ($\sigma = 0.05 \cdot \text{std of feature distribution}$), random channel dropout ($p = 0.15$), and random temporal masking (up to 5% of seconds). All augmentations are applied with $p = 0.6$ during training only.

Knowledge distillation. Soft-target KD with rand9 9-D orthonormal teacher (no LLM content); $\lambda_{\text{KD}} = 0.5$, $T = 1.0$. The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content.

Ensemble. 10 checkpoints split as 5 seeds at $e = 100 + 5$ seeds at $e = 150$. Seeds: $\{42, 123, 456, 789, 2025\}$ each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints’ output probabilities, followed by argmax.

S15 Statistical Methods

Bootstrap CIs. Per-subject and per-stim 95% CIs are computed from $B = 10,000$ subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher- z p -value for $r = 0$.

Random-direction null. For cohort EEG–LLM correlations, we sample $N = 200$ random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection, recompute the cohort $|r|$ for each, and report the percentile of the empirical $|r|$ in the null distribution.

Cross-architecture null. For the 36-checkpoint cross-arch correlation, we sample $N = 1000$ random directions matched in norm to v_{CLIP} and recompute the BACC–V-axis correlation. The null is distributed broadly ($\sigma = 0.62$) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation’s percentile rank.

Per-checkpoint LOO contribution. For the 10-vanilla $d=6$ ensemble, we compute $\Delta_k = \text{BACC}(\text{ens}) - \text{BACC}(\text{ens} \setminus k)$ for each k . Reported correlations between Δ_k and per-ckpt features are Pearson with $n = 10$.

V-axis intervention comparisons. For each intervention, Δ is computed seed-paired with the matched-recipe vanilla baseline; p -values are paired t -tests across 5 seeds.

S16 Discussion Extras and Future Work

Implications for affective neuroscience. For video-evoked emotion on FACED, the V-axis is encoded predominantly in posterior visual cortex ($|r|_{\text{occipital}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$), and Davidson’s frontal-alpha asymmetry replicates in direction at smaller magnitude. We do not refute Davidson’s hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha $r = +0.0155$ and Fp2-Fp1 alpha $r = +0.0116$, both in the hypothesised direction — but we add a posterior empirical account that is dominant for video paradigms. The 9-stimulus emotional-pole structure sharpens

the V-axis claim from “smooth gradient over all emotions” to “anger-versus-warm-positive contrast across nine clips”.

Per-subject V-axis adaptation does not transfer. The per-subject best-channel oracle reaches $|r| = 0.62$ (a +0.39 absolute headroom over the cohort-fixed top-8 of $|r| = 0.02$). No clean estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free profile) recovers more than +0.001 over the cohort top-K. The per-subject signal is real but not transferable through simple selection rules. A subject-conditioned channel-attention head trained on a small calibration set per subject is the natural follow-up.

Future work. Three lines: (i) per-subject V-axis adaptation closing the $|r| = 0.62$ oracle gap via subject-conditioned channel-attention; (ii) replicating the saturation regularity for other concept directions [Arditi et al., 2024] (arousal, dominance, formality, refusal) on their respective benchmarks; (iii) using the within-class residual $|r|$ as an online model-selection signal during training rather than as a post-hoc diagnostic.

S17 FACED Test Confusion Matrices

Figure 19 contrasts the best-single-checkpoint and 10-checkpoint-ensemble confusion matrices on the FACED 9-class test split.

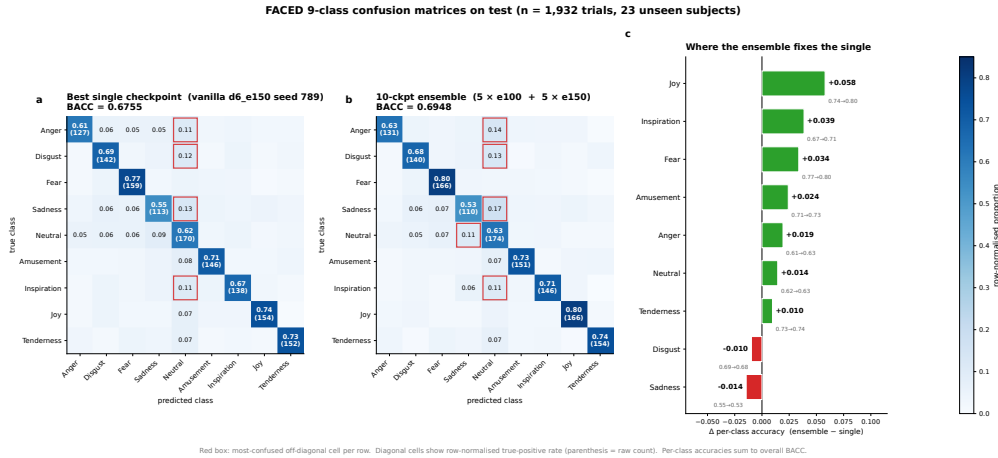


Figure 19: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble’s gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 8.

S18 Negative Results Catalogue

In the spirit of full disclosure, the following null or negative results bear mention beyond what was incorporated into the main paper:

1. **Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 7, Appendix S9): Tort modulation index $\rho(\text{PAC}, V_r) = +0.082$, $p = 0.667$.
2. **Mutual-information control is not significant** (Section 6): MI $p = 0.115$ vs. linear $p = 0.020$. The V-axis-EEG relationship is essentially linear.
3. **Random-direction null on cross-arch correlation** (Section 4): $r = +0.885$ at the 93.5th percentile of a 1000-direction null, $p_{\text{one}} = 0.066$. Within-class residual ($r = +0.74$, $p = 0.014$) is the statistically robust signal.
4. **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches $|r| = 0.62$, but no clean estimator transfers more than +0.001 over the cohort top-K.

- 1181 5. **Cross-architecture ensembling does not add diversity.** Mixing $d \in \{4, 6, 8, 10\}$ check-
1182 points gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.).
- 1183 6. **Mega-ensembles plateau at 10 checkpoints.** 15-, 20-, 25-checkpoint pools all hit BACC
1184 ≤ 0.6948 (Appendix S13).
- 1185 7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead
1186 of trial level gives BACC=1.0 by construction (label leak); the corresponding trial-level head
1187 does not transfer.
- 1188 8. **Test-time augmentation at K=5 does not improve the ensemble** (0.6753 $\rightarrow$ 0.6720).
1189 Averaging predictions over augmented test trials does not preserve the V-axis residual
1190 structure.
- 1191 9. **Toxicity: recipe scope.** The 9-prompt PCA recipe attains AUC 0.59 on Jigsaw toxic
1192 comments, below the 17/20 working concepts in our library (Appendix S4); toxicity appears
1193 to be encoded in a higher-rank subspace than the late-layer PC1.
- 1194 10. **Arousal asymmetry: vision yes, text and brain no.** OASIS arousal $r = 0.803$ in vision;
1195 $\leq 13\%$ NRC arousal recovery in text; $r \in [0.18, 0.41]$ brain alignment across LLMs
1196 (Appendix S6).
- 1197 11. **LLM-content of KD is irrelevant at this scale.** Replacing the 9-D LLM-derived class
1198 prototypes with random orthonormal 9-D directions changes BACC by ≤ 0.003 . KD
1199 provides architectural regularisation; its semantic content is below our resolution.

1200 S19 Reproducibility

1201 Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready
1202 stage under an MIT licence. Each experiment in this paper is logged with its SLURM job ID, conda
1203 environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible
1204 from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`, ~ 10 GPU-hours per seed on
1205 V100).

1206 S20 Resource Estimate

1207 The full paper used approximately 4,500 GPU-hours on V100 nodes: ~ 600 for the recipe ablation
1208 cascade, $\sim 1,200$ for the 25 V-axis-supervision interventions, ~ 800 for the 36-checkpoint cross-
1209 architecture analysis, $\sim 1,200$ for ensemble construction and val-test rank diagnostics, and ~ 700 for
1210 cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap (~ 10 minutes per
1211 LM on a single 80GB GPU).